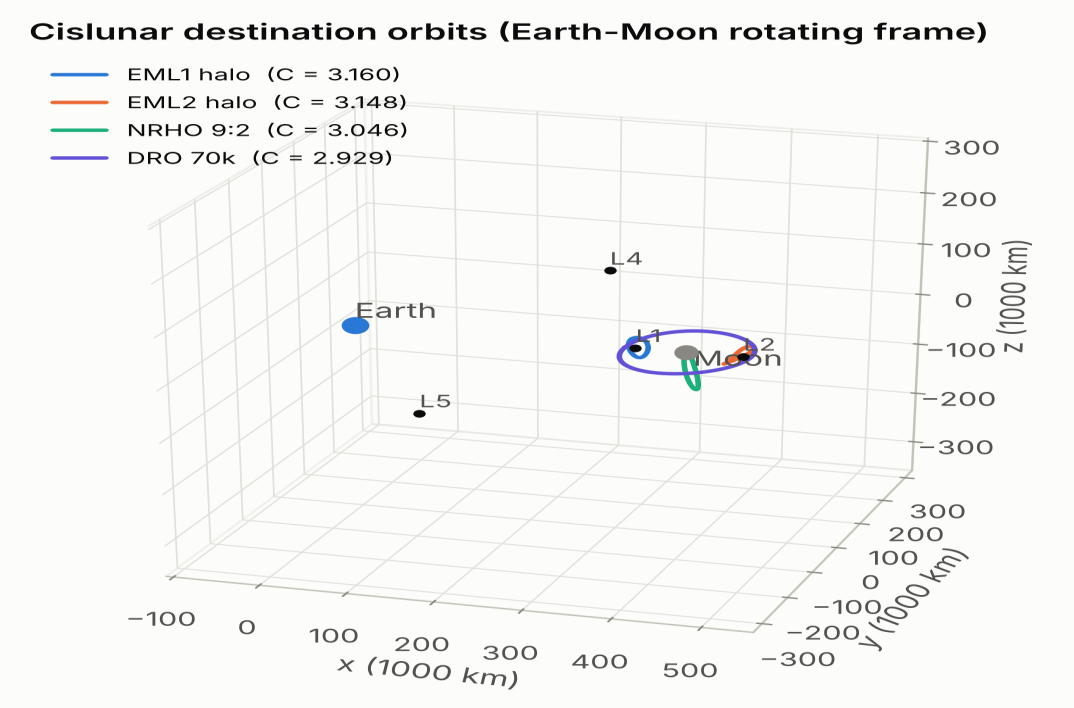
CISLUNAR SOLAR SAILING FOR DARPA CATTLE RUN

Mission Simulation Study: Results Report

Transit times, launch geometry, cislunar transfers, pole-sitting, 'in-irons' limits, field of view, hybrid manoeuvrability and the lowest acceptable demonstration orbit for distributed solar sails



Prepared for	Cattle Run proposal team (DARPA-SN-27-03, Proposers Day 10 November 2026)
Date	1 November 2026
Basis	Objectives & outline (ObjectivesAndOutline.docx); reference library in Resources/; open-source simulator in simulator/ (this repository)
Deliverables	This report (PDF) and the companion briefing (PPTX); all figures, results JSON and code are reproducible with **python simulator/run_all.py**

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Executive summary

DARPA's **Cattle Run** program (Special Notice DARPA-SN-27-03) seeks cost-effective cislunar constellations built from autonomous spacecraft that use solar sails or other renewable propulsion for station keeping and manoeuvring over long missions. The first step is a demonstrator launched *below GEO* to show sufficient acceleration, accurate vehicle control and autonomous GNC; later steps fly multiple spacecraft in cislunar space. This study built an end-to-end, GPU-capable simulator from the project outline and used it, together with the reference library, to answer the eight study questions. Results are given for a Solar Cruiser-class conventional sail (**CR-S**: 1,653 m^2, 111 kg, a_c = 0.123 mm/s^2), a hybrid with a 20 mN gimballed SEP module (**CR-H**: a_c = 0.104 mm/s^2 + SEP 0.153 mm/s^2) and parametric sweeps of characteristic acceleration.

#	Question	Answer (headline numbers)
1	Transit time from GTO, GEO, TLI, LEO to cislunar space	CR-S reaches lunar distance in **490 d** from GEO, **818 d** from a perigee-raised GTO and **1,991 d** from 1,000 km LEO; times scale as $1/a_c$ (66 d from GEO at 1 mm/s^2). TLI reaches lunar distance in 2.6 d; sail capture to halo energy takes ~89 d (hybrid 27 d).
2	Optimal launch inclination	Match the lunar orbit plane at arrival: 26.6 deg (2028) falling to 23.5 deg (2030). A standard 28.5 deg Cape Canaveral launch / 27 deg GTO is within 1 % of optimal; SSO launches cost ~130-140 extra days.
3	Transit to/between DRO, LRO-class LLO, EML1/2/4/5	EML1/EML2 halos, NRHO and DRO are mutually reachable in 27-77 d with the sail (18-38 d hybrid); L4/L5 in 60-79 d; any leg to or from a 100 km LLO takes 298-367 d, dominated by a 293 d lunar spiral (hybrid 78-137 d).
4	Lunar pole-sitter requirements	Static hovering 60,000-100,000 km below the south pole needs 1.15-1.77 mm/s^2 continuously (36-56 km/s per year). A sail can never hold it alone; synodic-period coverage orbits need a_c >= 0.58-1.7 mm/s^2 (5-14x CR-S).
5	'No travel' / 'in irons' region	Al/CP1 sail: zero thrust within 34.5 deg of the Sun direction, <5 % a_c within 50.8 deg; in irons 28 % of the time in orbit raising and 18 % for random station-keeping demands. Every hybrid variant removes it.
6	Field of view: fixed vs gimballed bus	Propulsion-driven attitude puts a body-fixed 10 deg instrument on Earth only 3 % of the time; a +/-60 deg two-axis gimbal raises it to 39 % (65 % for anti-Sun SDA fields) and solar power from 75 % to 99 %.
7	Hybrid manoeuvrability and acceleration	CR-H: 0.257/0.181/0.116 mm/s^2 (away/perpendicular/toward Sun) vs 0.123/0.043/0 for CR-S; 7-day reach from a DRO 49k km vs 22k km; hybrid is lighter than SEP-only above 6.7 km/s cumulative delta-v.
8	Lowest acceptable orbit for the demonstrator	Drag equals sail thrust at 505/655/785 km (solar min/moderate/max). A robust demo needs **~1,000 km** (900 km in dawn-dusk SSO); from a 550 km rideshare a stowed-sail SEP climb takes 18 d and 2.1 kg.

Recommendations

1. Baseline a hybrid sail. A Solar Cruiser-class sail with a small two-axis gimballed SEP module (CR-H) removes the in-irons region, doubles manoeuvre reach, and decouples power and pointing from propulsion at a 20 kg (18 %) mass cost.

2. Fly the below-GEO demonstrator at ~1,000 km in a dawn-dusk sun-synchronous orbit (no eclipses, ~20 % faster raising than noon-midnight, drag < 25 % of sail thrust even at solar maximum). If only a 550 km rideshare is available, climb with the sail stowed using SEP before deployment.

3. Launch cislunar elements from Cape Canaveral (28.5 deg) or into a 27 deg GTO; raise perigee to ~1,000 km before sail deployment (75 m/s) and use DRO, NRHO and L1/L2 halos as constellation slots; reach L4/L5 within 2-3 months.

4. Do not baseline a lunar pole-sitter. Provide south-pole coverage with two spacecraft in NRHO/L2-halo orbits; reserve pole-sitting for a far-term sail (a_c >= 1 mm/s^2) or a short SEP-dominated campaign.

5. Gimbal the bus (or payload) relative to the sail and carry antennas on both faces; use reflectivity-control devices plus centre-of-mass shift for propellant-free momentum management.

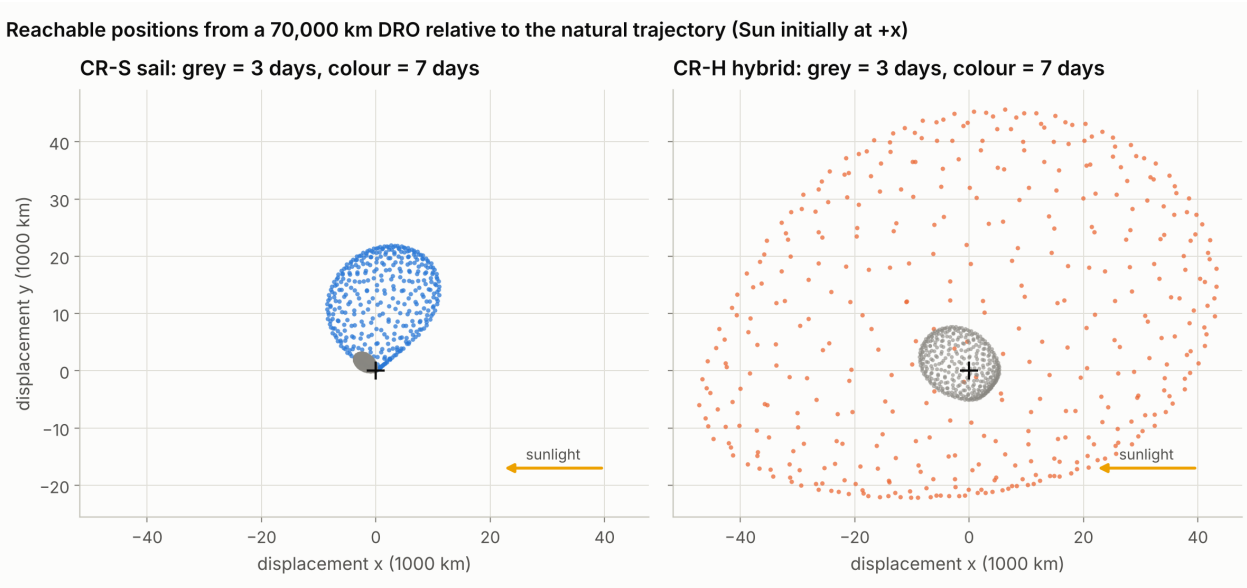


Figure ES-1 - Why the hybrid: positions reachable from a 70,000 km DRO in 3 days (grey) and 7 days (colour). The sail's reach is lopsided away from the Sun; the hybrid reaches every direction.

1. Program context and study objectives

The Cattle Run Proposers Day (10 November 2026) frames a two-phase program: a rapid demonstrator below geosynchronous orbit, then multiple spacecraft operating in cislunar space. The project objectives document restates these as four mission objectives and asks for an inclusive simulator that covers the trip from Earth to and within cislunar space, estimates transit times, contrasts conventional and hybrid sails, uses PyTorch to exploit GPUs, pre-processes mission profiles for the design options and produces 2-D and 3-D plots for presentations.

Mission objective (DARPA / outline)	How this study addresses it
Cost-effective cislunar constellation	Orbit slots (DRO, NRHO, L1/L2 halos, L4/L5), transfer-time matrix, deployment and phasing times (Sections 4-5)
Autonomous spacecraft using solar sails or other renewable propulsion	Locally optimal on-board steering laws (closed form / table driven) used throughout; hybrid SEP allocation
Station keeping and manoeuvrability over long durations	In-irons statistics (Q5), reachable sets and response times (Q7), propellant-free momentum management
Launch below GEO to show acceleration, control and autonomous GNC	Drag, gravity-gradient, atomic-oxygen and solar-cycle analysis of candidate orbits (Q8)

The design options and variables listed in the outline - conventional vs hybrid sail, fixed vs gimballed SEP, repeatedly deployable/retractable sail, embedded electrochromic films, single- vs double-sided film, mass, sail material and size, SEP thrust and specific impulse, rigid booms vs spin deployment and gimbal range - are all parameters of the simulator and are traded in Section 12.

2. Simulation framework

The simulator (**simulator/cislunar_sail**) is a Python package with an array-backend abstraction: every batched propagator and steering law runs unchanged on NumPy or on PyTorch tensors (CPU or CUDA GPU; select with **CISLUNAR_BACKEND=torch**). Thousands of trajectories - Sun phases, launch seasons, design points, steering commands - are propagated in lock-step, which is how the Monte-Carlo-style ensembles in this report were produced.

Module	Content
sail	Optical force model (Eq. 1-2), material catalogue, locally optimal cone-angle tables, force envelopes, sail+SEP allocation with gimbal limits and drag-aware steering, membrane temperature
environment	Sun and Moon ephemerides (18.6-yr nodal cycle), cylindrical Earth/Moon shadows, NRLMSIS 2.x density tables vs altitude, local time and solar activity (F10.7 = 70/150/230)
earth_orbit	Orbit-averaged Gauss equations in modified equinoctial elements (non-singular) with J2, eclipses, drag, SEP mass flow; batched RK4; Moon-centred variant; Cowell integrator with Sun + Moon gravity for validation
cr3bp	Earth-Moon CR3BP with rotating Sun line (non-autonomous), optional solar tidal gravity (BCR4BP), shadows, differential correction and continuation for Lyapunov, halo, NRHO and DRO families, batched RK4
designs	Heritage and study design points; parametric sail-system mass model (membrane, booms, deployer, RCD, retraction) calibrated on ACS3 and Solar Cruiser; SEP sizing after Ceriotti & McInnes
analyses	One module per study question (q1-q8) plus design trades; run_all.py regenerates every JSON and figure

2.1 Physical models

The sail force uses the McInnes non-ideal optical model with the Rios-Reyes & Scheeres coefficients. **s** is the photon direction (Sun to sail), **n** the normal of the lit face, alpha the cone angle:

$$\mathbf{F} = PA\cos\alpha\left[a_3\,\hat{\mathbf{s}} + (a_1\cos\alpha + a_2)\,\hat{\mathbf{n}}\right], \qquad P = W/c = 4.54\ \mu\mathrm{N\,m^{-2}}$$

$$a_1 = 2rs, \quad a_2 = B_f(1-s)\,r + (1-r)\,\frac{\varepsilon_f B_f - \varepsilon_b B_b}{\varepsilon_f + \varepsilon_b}, \quad a_3 = 1 - rs, \qquad a_c = \frac{PA}{m}\,(a_1 + a_2 + a_3)$$

Characteristic acceleration a_c is always quoted for the real (non-ideal) film at 1 AU with the lit face normal to the Sun; this is ~9 % below the ideal 2PA/m formula used in some flight-project quotes (e.g. ACS3's 0.045 mm/s^2). In the Earth-Moon rotating frame (Earth at -mu, Moon at 1-mu) the dynamics and the rate of change of the Jacobi constant C, used to measure energy changes between orbits, are:

$$\ddot{\mathbf{r}} + 2\boldsymbol{\omega}\times\dot{\mathbf{r}} = -\nabla U(\mathbf{r}) + \mathbf{a}_{sail} + \mathbf{a}_{SEP}, \qquad \frac{dC}{dt} = -2\,\dot{\mathbf{r}}\cdot(\mathbf{a}_{sail} + \mathbf{a}_{SEP})$$

Steering is locally optimal: for any desired direction (velocity for energy, the gradient of apogee radius for apogee raising, an arbitrary control demand for station keeping) the cone angle that maximises the useful force component is taken from a precomputed table; the sail feathers edge-on when nothing useful is available. For hybrids the SEP thrust may point within the gimbal range of either sail-normal direction, and a brute-force search over cone angle finds the best combined component, optionally penalising drag.

2.2 Validation against the reference library

Benchmark	Reference value	Simulator
Optimal cone angle / thrust fraction for thrust perpendicular to sunlight (ideal sail)	35.26 deg / 0.385 (McInnes 1999)	35.25 deg / 0.385
Orbit-averaged transverse thrust fraction, Sun in plane (ideal sail)	0.4393 (McInnes Sec. 4.4.4)	0.4393
Spiral from GEO to 326,000 km at a_c = 0.5/0.2/0.1 mm/s^2	104/259/519 d (quasi-circular estimate)	95/243/492 d (eccentricity grows)
ACS3 SMA rise in 180 days, 715 km / 900 km SSO (Dono 2023)	+301 km / +340 km	+212 to +338 km / +353 to +371 km (F10.7 150 to 70)
Minimum altitude for ACS3 orbit raising (Wilkie 2023)	~691 km (mean activity)	647 km noon-midnight SSO (F10.7 = 150)
Averaged vs Cowell (Sun+Moon gravity), raised GTO, 60 days	-	apogee 58,854 vs 58,779 km (0.1 %)
Earth-Moon L1/L2 Jacobi constants	3.1883 / 3.1722	3.18834 / 3.17216
9:2 NRHO (Gateway reference)	x0 1.0221, z0 -0.1821, vy0 -0.1033, 6.56 d, perilune ~3,250 km	1.02202, -0.18210, -0.10326, 6.56 d, 3,249 km
Hover acceleration below the lunar south pole (Ozimek, Grebow & Howell)	contours 0.58-1.7 mm/s^2 near L1/L2	map reproduced (Fig. 4.1); 1.77 mm/s^2 at 60,000 km

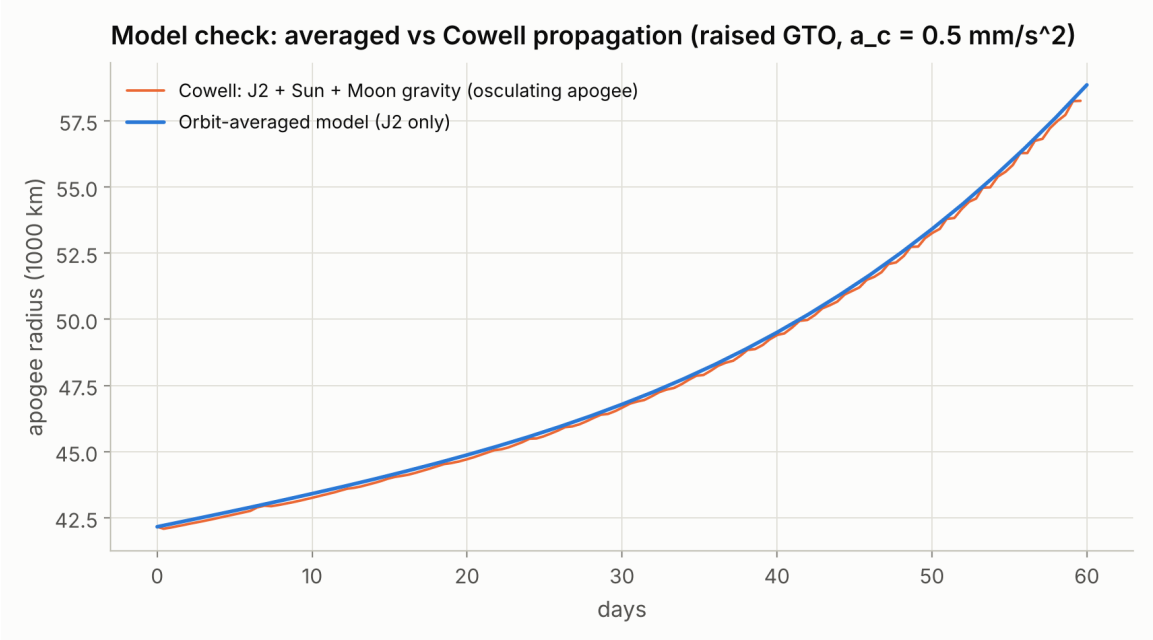


Figure 2.1 - Orbit-averaged propagation (J2 only) against direct Cowell integration with J2, solar and lunar gravity for a perigee-raised GTO spiral at a_c = 0.5 mm/s^2.

3. Spacecraft configurations and design points

Heritage sailcraft anchor the performance range; three study designs bracket the Cattle Run options. CR-S reuses the Solar Cruiser sail system (TRL-6 quadrant tested 2022-2024); CR-H adds a 20 kg SEP module (20 mN Hall thruster at ~350 W, Isp 1,500 s, 6 kg xenon/krypton, two-axis +/-30 deg gimbal); CR-HR adds a motorised re-spooling deployer; CR-A is a far-term 4,000 m^2 sail on 1 um film.

Design	Mass (kg)	Area (m^2)	Film	a_c (mm/s^2)	Loading (g/m^2)	SEP	SEP accel. (mm/s^2)	SEP dv (km/s)
LightSail 2 (2019)	5	32	Al/Mylar 4.5 um	0.053	156.2	-	-	-
ACS3 (2024)	16	80	Al/PEN/Cr 2.1 um	0.041	200.0	-	-	-
NEA Scout (2022)	14	86	Al/CP1 2.5 um	0.051	162.8	-	-	-
Solar Cruiser (TRL-6 sail)	111	1,653	Al/CP1 2.5 um	0.123	67.2	-	-	-
Cattle Run conventional sail	111	1,653	Al/CP1 2.5 um	0.123	67.2	-	-	-
Cattle Run hybrid (gimballed SEP)	131	1,653	Al/CP1 2.5 um	0.104	79.2	20 mN, 1500 s, +/-30 deg	0.153	0.69
Cattle Run hybrid, retractable sail	136	1,653	Al/CP1 2.5 um	0.100	82.3	20 mN, 1500 s, +/-30 deg	0.147	0.66
Advanced sail (DCB-II, 1 um CP1)	120	4,000	Al/CP1 1 um	0.275	30.0	-	-	-

Film / model	r	s	eps_f / eps_b	Thrust efficiency	Max force angle (deg)	No-go cone (deg)	T at normal incidence
Ideal	1	1	-	1.000	~90	~0	-
Al/CP1 2.5 um (Solar Cruiser, NEA Scout)	0.88	0.94	0.05 / 0.55	0.908	55.5	34.5	-10 C
Al/PEN/Cr 2.1 um (ACS3)	0.90	0.82	0.03 / 0.60	0.893	49.2	40.8	-25 C
Al/CP1/Al double-sided	0.88	0.94	0.05 / 0.05	0.934	57.1	32.9	139 C

Table 3.1/3.2 - Design points (non-ideal a_c) and film optical properties (McInnes 1999 Table 2.1; ACS3 appendix in Wilkie 2023).

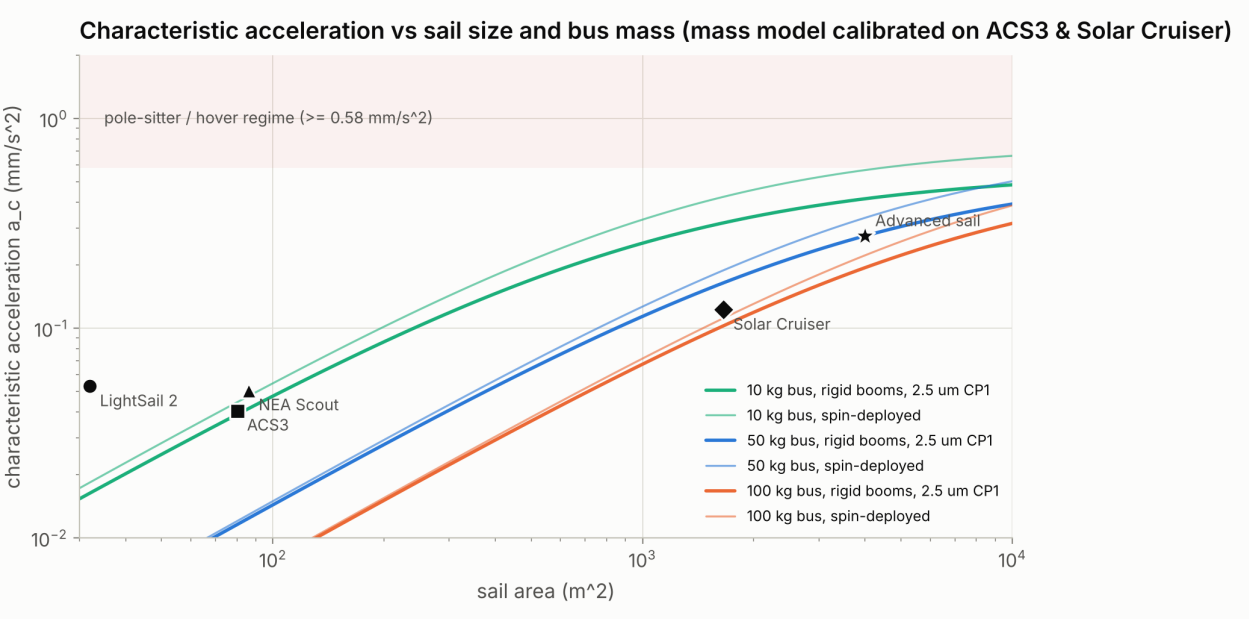


Figure 3.1 - Characteristic acceleration against sail area for 10/50/100 kg bus-plus-payload masses, rigid booms (thick) and spin deployment (thin), from the calibrated mass model, with heritage points.

4. Q1 - Transit time from GTO, GEO, TLI and LEO to cislunar space

Method. Batched orbit-averaged propagation with locally optimal apogee-raising steering and perigee protection, eclipses and J2; six launch seasons per case (Sun geometry), arrival when the osculating apogee reaches 326,000 km (the Earth-Moon L1 distance). The L1 energy 'gateway' (Jacobi constant C = C_{L1}) is already crossed at r_a ~ 260,000 km, after which lunar gravity and sail steering complete the capture (Section 6). Hybrid cases use CR-H with its 6 kg of propellant and, for reference, unlimited propellant.

Start orbit	0.045 (ACS3)	0.07	0.10	0.123 (CR-S)	0.2	0.3	0.5	1.0	CR-H 6 kg	CR-H unlimited
LEO 1,000 km (28.5 deg)	n/a	3,569	2,478	**1,991**	1,248	822	495	242	1,980	430 (46 kg)
GTO, perigee raised to 1,000 km	2,306	1,411	990	**818**	508	354	216	106	756	206 (23 kg)
GEO (35,786 km)	1,434	868	643	**490**	340	196	120	66	395	122 (14 kg)

Table 4.1 - Median days to lunar-distance apogee vs characteristic acceleration (mm/s^2); 'n/a' = longer than 10 years.

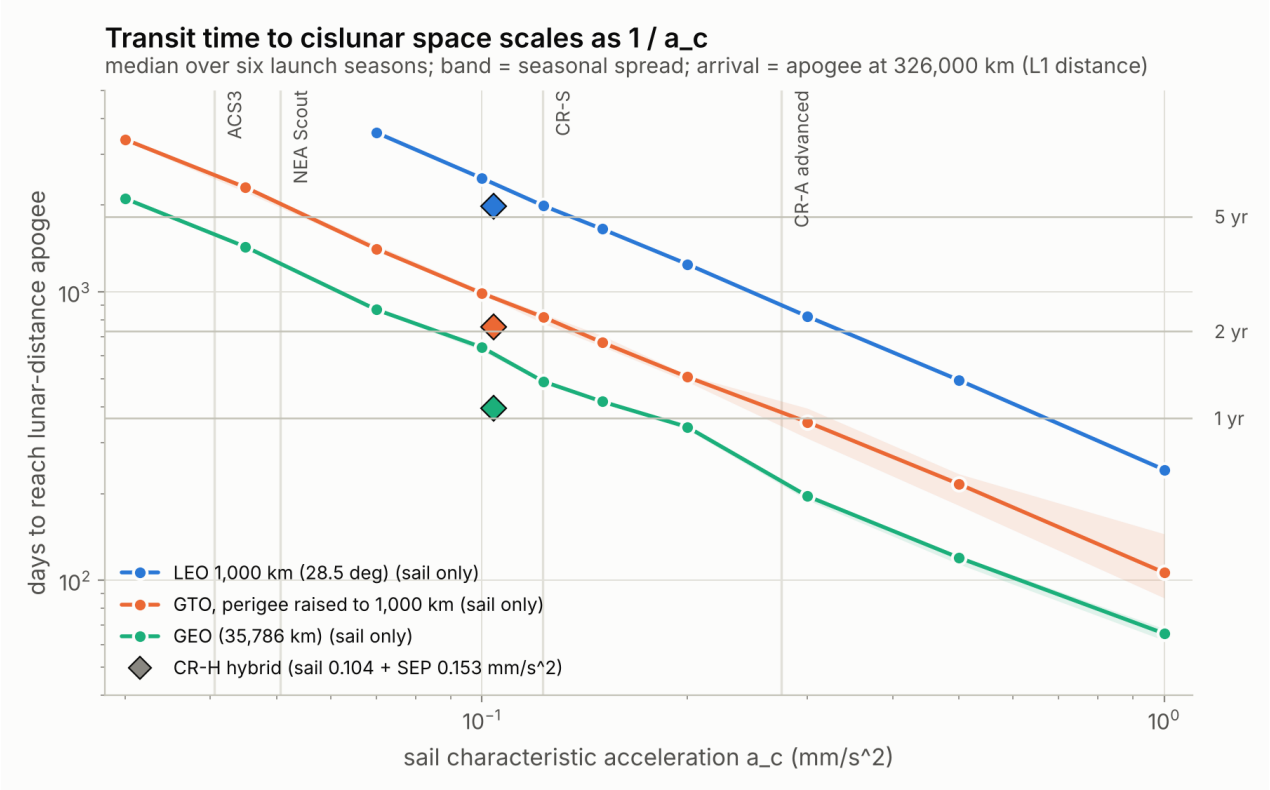


Figure 4.1 - Transit time to lunar distance against a_c for three start orbits (bands: spread over launch seasons) and the CR-H hybrid (diamonds).

Findings

- Transit time is essentially Delta-v / (useful fraction x a_c): from GEO the spiral needs ~2.1 km/s and the sail delivers ~0.4 a_c on average, so a_c = 0.1 mm/s^2 takes ~1.8 years.
- Starting low is expensive: 1,000 km LEO takes 5.5 years for CR-S versus 1.3 years from GEO. LEO also adds drag, eclipses and atomic oxygen (Section 11), so LEO is a demonstration orbit, not a staging orbit.
- Standard GTO (250 km perigee) is survivable only if the sail holds an edge-on attitude through every 10 km/s perigee pass: with ~1 % residual drag area the drag-aware law lifts perigee to 3,450 km in 120 days for CR-S. The robust option is a 75 m/s perigee raise to 1,000 km before deployment.
- TLI places the spacecraft at lunar distance in 2.6 days; the sail must then shed energy to be captured (C from 2.27 to the EML2-halo value 3.15): median 89 d for CR-S and 27 d for CR-H. Ballistic lunar transfers (CAPSTONE, ~4 months) reach the same orbits with almost no propulsion and are an attractive alternative.
- The hybrid's 6 kg of propellant shortens transits only modestly (CR-H is 18 % heavier, so its sail is weaker); with unlimited SEP it reaches lunar distance from GEO in 122 d but burns 14 kg - transits are where SEP propellant is least valuable.

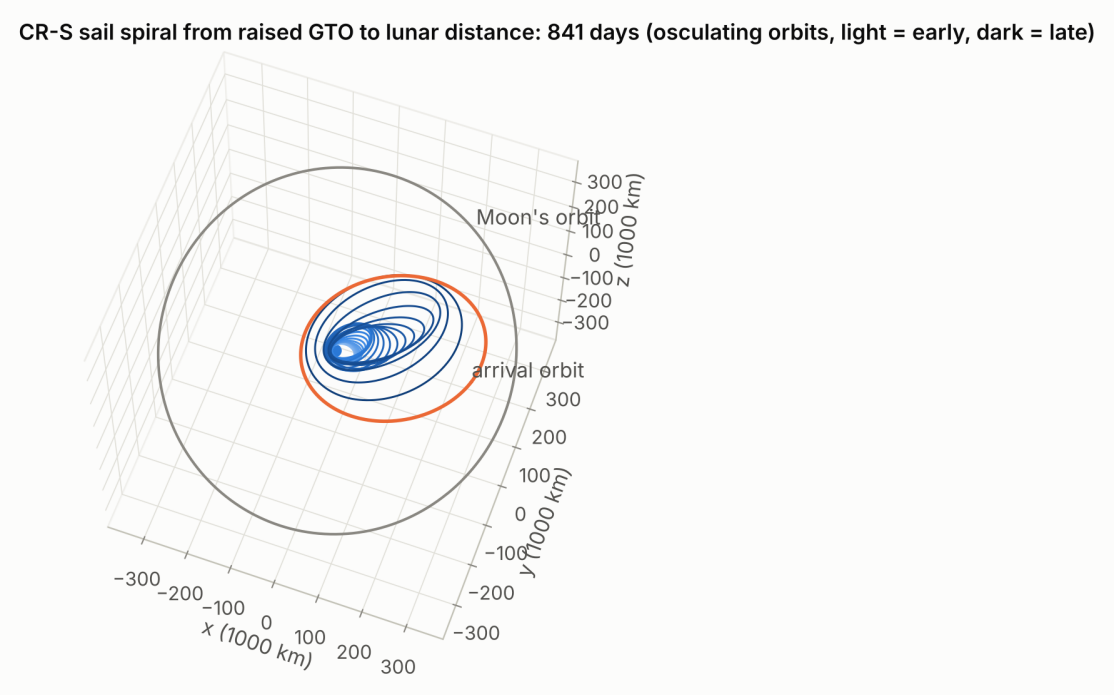


Figure 4.2 - 3-D view of the CR-S spiral from a perigee-raised GTO: osculating orbits coloured light (early) to dark (late), with the Moon's orbit.

Answer to Q1

For a near-term sail (a_c ~ 0.12 mm/s^2): **~1.3 yr from GEO, ~2.2 yr from a perigee-raised GTO, ~5.5 yr from 1,000 km LEO**; from TLI, lunar distance in 2.6 d plus ~89 d sail capture. Doubling a_c halves every number. Cislunar elements should therefore ride to GTO/GEO or TLI and use the sail for the final leg and for operations, not for the climb out of LEO.

5. Q2 - Optimal launch inclination

Three effects set the best inclination: the lunar orbit plane, whose inclination to Earth's equator swings between 18.3 and 28.6 deg with the 18.6-year nodal cycle; sail effectiveness, which depends on the Sun's elevation beta above the orbit plane and on eclipses; and the cost of removing any plane mismatch. We swept launch inclinations from 0 to 97.6 deg for perigee-raised GTO spirals (six seasons each) and measured plane-change rates on the high-apogee arrival orbit.

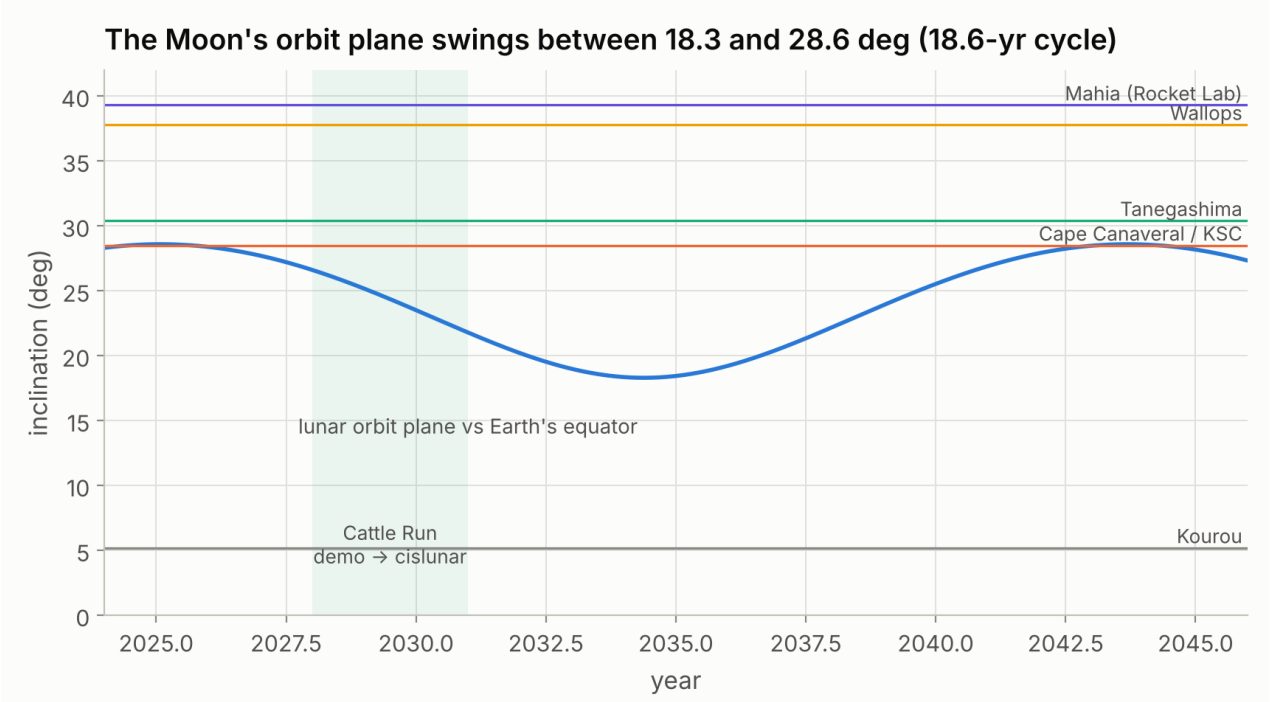


Figure 5.1 - Inclination of the lunar orbit to the equator, 2024-2046, with launch-site latitudes. The 2025 major lunar standstill makes 2027-2030 well matched to Cape Canaveral launches.

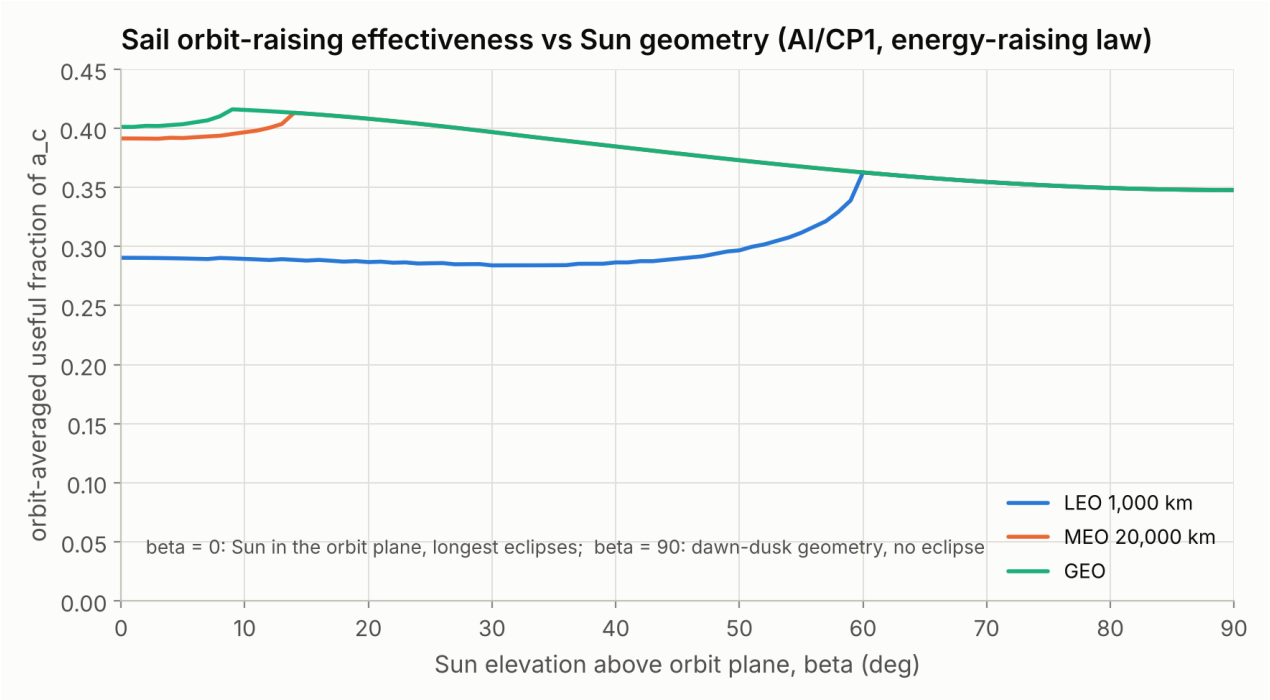


Figure 5.2 - Orbit-averaged useful thrust fraction for energy raising vs Sun elevation above the orbit plane (AI/CP1).

In LEO the dawn-dusk geometry (beta = 90 deg) is best (0.348 vs 0.290 of a_c) because a third of a beta = 0 orbit is in eclipse (33 %); at GEO the Sun in the plane is best (0.401). Over a multi-month spiral the seasons average out, so the spiral time itself varies by only +/-3 % with inclination.

Launch inclination (deg)	0	10	18.3	23.4	28.5	35	45	51.6	63.4	90	97.6
Spiral to lunar distance (d)	821	822	812	815	822	826	832	816	797	824	824
Inclination on arrival (deg)	4.4	9.2	14.0	17.6	21.1	27.2	42.4	54.9	66.5	90.8	96.7
Total incl. plane match (d)	857	849	830	826	826	834	868	877	880	953	964

Table 5.1 - CR-S from a perigee-raised GTO (launch 2028, arrival ~2030.3; lunar plane at arrival 23.1 deg). Plane change near apogee costs ~1.6 d/deg with apsides aligned to the node line (~38 d/deg if perpendicular).

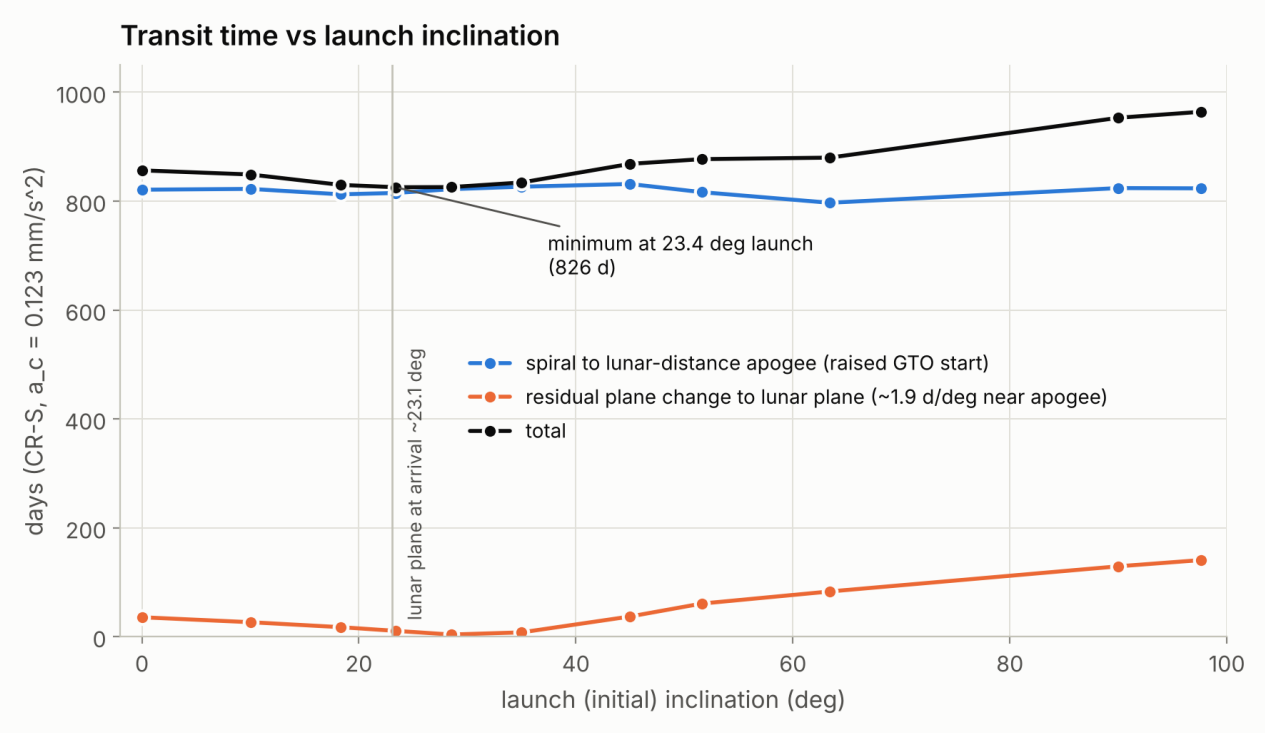


Figure 5.3 - Spiral time, residual plane change and total vs launch inclination.

A useful by-product: the out-of-plane part of the sail force drifts the orbit plane toward the ecliptic (23.4 deg), which lies within 5.1 deg of the lunar plane - a 28.5 deg launch arrives at ~21 deg. RAAN need not be matched at launch: J2 regresses the node of a GTO by ~0.3 deg/day, so the arrival node is set by launch date and spiral duration.

Answer to Q2

Launch close to the lunar-plane inclination at arrival: ~23-29 deg for 2028-2031. The standard Cape Canaveral 28.5 deg launch (or a 27 deg GTO rideshare) is within 1 % of the optimum (826 d total for CR-S). Equatorial launches cost ~31 d more and sun-synchronous launches ~139 d more. For later constellation launches (2032-2036) the lunar plane drops to 18.3 deg, favouring ~20 deg launches. For the LEO demonstrator, choose a dawn-dusk sun-synchronous orbit.

6. Q3 - Transit to and between DRO, LRO-class orbit and EML1/2/4/5

Destinations. Periodic orbits were computed by differential correction and continuation in the CR3BP: an EML1 northern halo (Az 15,000 km), an EML2 halo (Az 10,000 km), the 9:2 NRHO, a 70,000 km DRO, the triangular points L4/L5, and a 100 km polar low lunar orbit representative of the Lunar Reconnaissance Orbiter (LRO flew 50 km circular, now ~20 x 165 km).

Orbit	Jacobi C	Period (d)	Stability index	Closest / farthest from Moon (km)
EML1 halo	3.1596	12.0	911.9	49,797 / 65,183
EML2 halo	3.1477	14.8	561.5	50,259 / 74,819
NRHO 9:2	3.0465	6.6	1.3	3,296 / 71,220
DRO 70k	2.9285	13.9	1.0	70,000 / 95,267
EML4	2.9880	-	1.0	-
LLO 100 km polar	5.4935	0.1	1.0	-

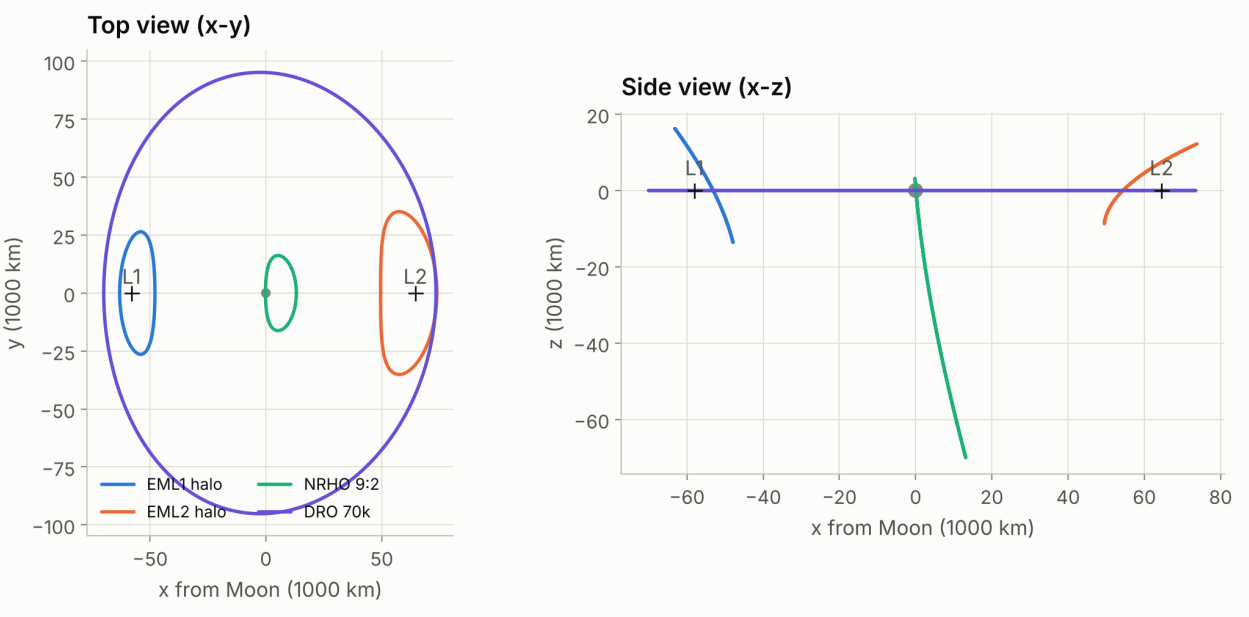


Figure 6.1 - Destination orbits seen from above and from the side (Moon-centred, rotating frame).

Method. Transfer time = time to change the Jacobi constant from the origin to the destination value under the locally optimal energy law (CR3BP with the Sun line rotating at the synodic rate; median over 12 starting phases x 8 Sun phases) + a natural coast between the regions once energies match. Coasts: L1-L2 21 d (computed along the L1-halo unstable manifold), halo/NRHO-DRO 20 d, Moon region-L4/L5 45 d, L4-L5 90 d. Legs to or from the 100 km LLO use a Moon-centred averaged spiral to a 20,000 km hand-over orbit (whose energy is close to the L2-halo value) plus the remaining energy change.

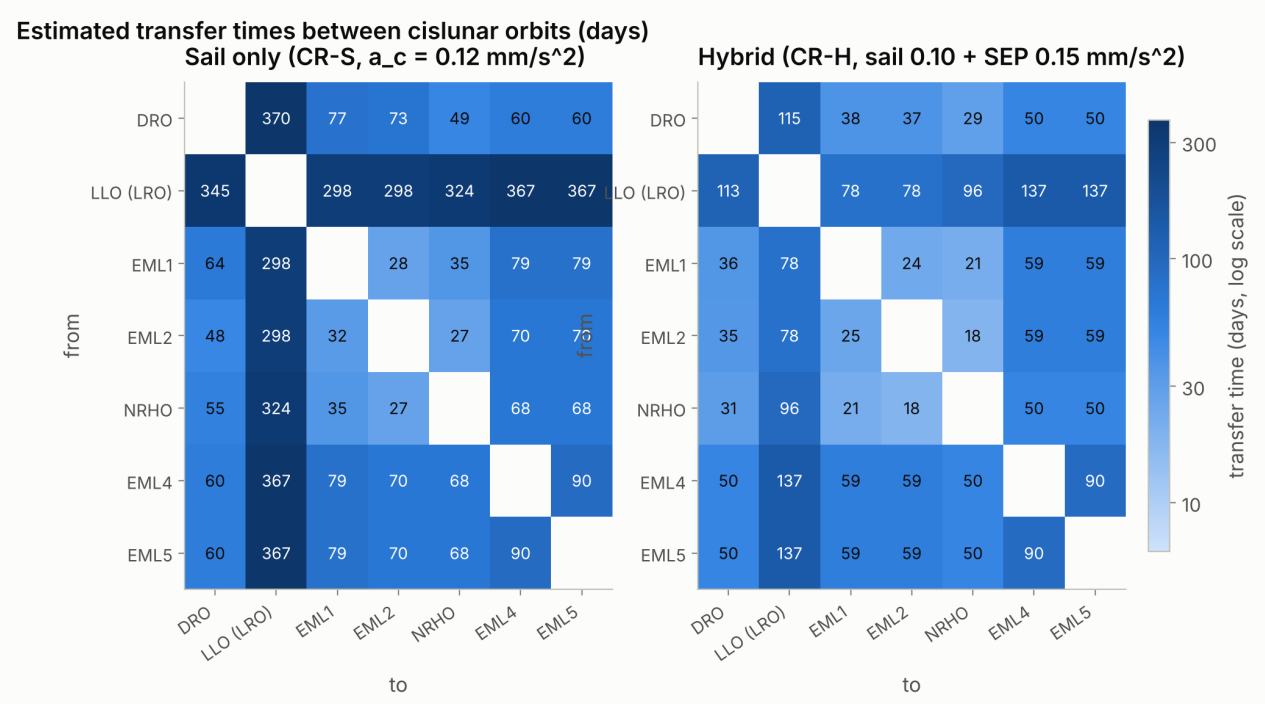


Figure 6.2 - Estimated transfer times (days) between destination orbits for the CR-S sail and the CR-H hybrid (row = from, column = to).

From \ to	DRO	LLO (LRO)	EML1	EML2	NRHO	EML4	EML5
DRO	-	370 / 115	77 / 38	73 / 37	49 / 29	60 / 50	60 / 50
LLO (LRO)	345 / 113	-	298 / 78	298 / 78	324 / 96	367 / 137	367 / 137
EML1	64 / 36	298 / 78	-	28 / 24	35 / 21	79 / 59	79 / 59
EML2	48 / 35	298 / 78	32 / 25	-	27 / 18	70 / 59	70 / 59
NRHO	55 / 31	324 / 96	35 / 21	27 / 18	-	68 / 50	68 / 50
EML4	60 / 50	367 / 137	79 / 59	70 / 59	68 / 50	-	90 / 90
EML5	60 / 50	367 / 137	79 / 59	70 / 59	68 / 50	90 / 90	-

Table 6.2 - Transfer times in days, sail (CR-S) / hybrid (CR-H).

- Halo, NRHO and DRO slots are mutually accessible in 27-77 days: their Jacobi constants differ by <= 0.23 and a 0.12 mm/s^2 sail changes C by 0.003-0.008 per day near these orbits.
- L4/L5 sit only Delta-C ~ 0.16 above the L2 halo energy but ~380,000 km away; the drift dominates (70 d sail, 59 d hybrid from the L2 halo).
- Low lunar orbit is the outlier: a 100 km orbit is deep in the lunar well (C = 5.49) and the sail spiral to 20,000 km takes 293 d with eclipses on every revolution; the hybrid needs 73 d but would spend far more propellant than CR-H carries.
- Departure timing matters: the fastest quarter of Sun and start phases is 6-24 % quicker than the median, and 0-23 % of phases fail within 130 days because the trajectory escapes or strikes the Moon first (medians are over successful cases). Lowering energy from the NRHO drives perilune into the Moon, so NRHO-to-halo legs need perilune control; they are estimated from the reverse legs.

- The hybrid shortens halo/NRHO/DRO legs by 15-51 %, L4/L5 legs by 16-26 % (the long drift is unchanged) and LLO legs by 74 % - but the 73-day hybrid lunar spiral needs ~0.6 km/s of SEP, most of CR-H's 6 kg load.

Answer to Q3

With a near-term sail (CR-S): **EML1 <-> EML2 ~28-32 d, halo/NRHO <-> DRO 48-77 d, to L4/L5 60-79 d, and any leg touching a 100 km LLO 298-367 d**. From TLI, sail capture to halo energy takes ~89 d (51 % of arrival geometries succeed) versus ~27 d for the hybrid. The hybrid shortens high cislunar legs by 15-51 %. LRO-class orbits should be served by dedicated SEP or chemical vehicles; the sail constellation belongs in the high cislunar orbits.

7. Q4 - What is necessary to achieve a lunar pole-sitter orbit?

A pole-sitter stays fixed below (or above) a lunar pole in the rotating frame. Its control acceleration must cancel gravity plus centrifugal acceleration at every instant. A sail can only produce vectors on its force 'bubble', which turns with the Sun once per synodic month, so we compute the residual that SEP must supply:

$$\mathbf{a}_{req}(\mathbf{r}) = \nabla U(\mathbf{r}), \qquad a_{SEP}(t) = \min_{\hat{\mathbf{n}}} \left| \mathbf{a}_{req} - \mathbf{a}_{sail}(\hat{\mathbf{n}}, \hat{\mathbf{s}}(t)) \right|$$

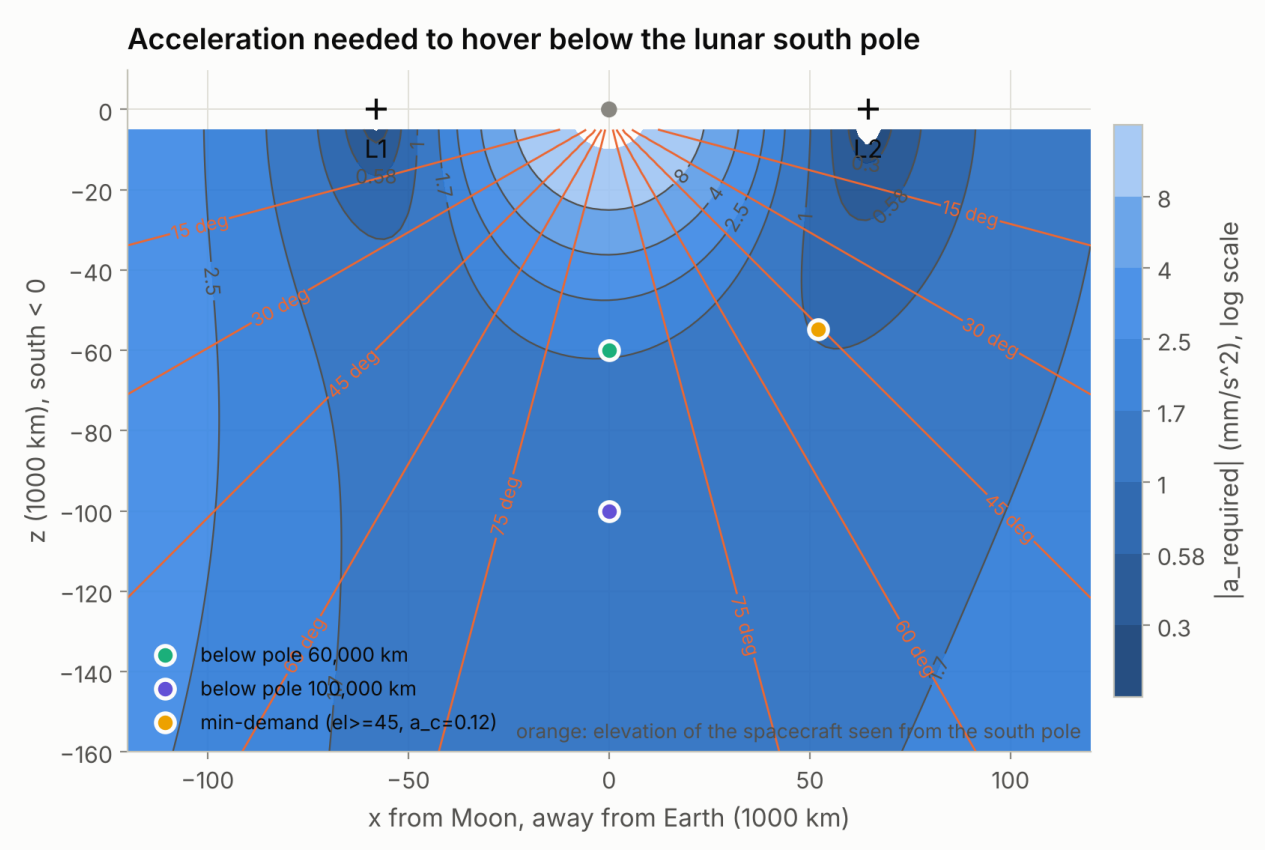


Figure 7.1 - Required hover acceleration below the lunar south pole (log colour scale; contours in mm/s^2) with elevation angles seen from the pole (orange) and candidate stations.

Station (Moon-centred)	|a_req| (mm/s^2)	Pure SEP dv/yr (km/s)	Prop. fraction / yr (Isp 3,000 s)	With sail 0.12	With sail 1.7	Sail alone holds hover
60,000 km below pole (el. 90 deg)	1.77	55.9	85 %	54.6 km/s	43.6 km/s	0 % of month
100,000 km below pole	1.15	36.4	71 %	35.1 km/s	26.1 km/s	0 % of month
best point with el. >= 45 deg	0.98	30.9	65 %	29.6 km/s	20.9 km/s	0 % of month

Table 7.1 - Pole-sitter budgets: SEP delta-v per year still required with sails of 0.12 and 1.7 mm/s^2.

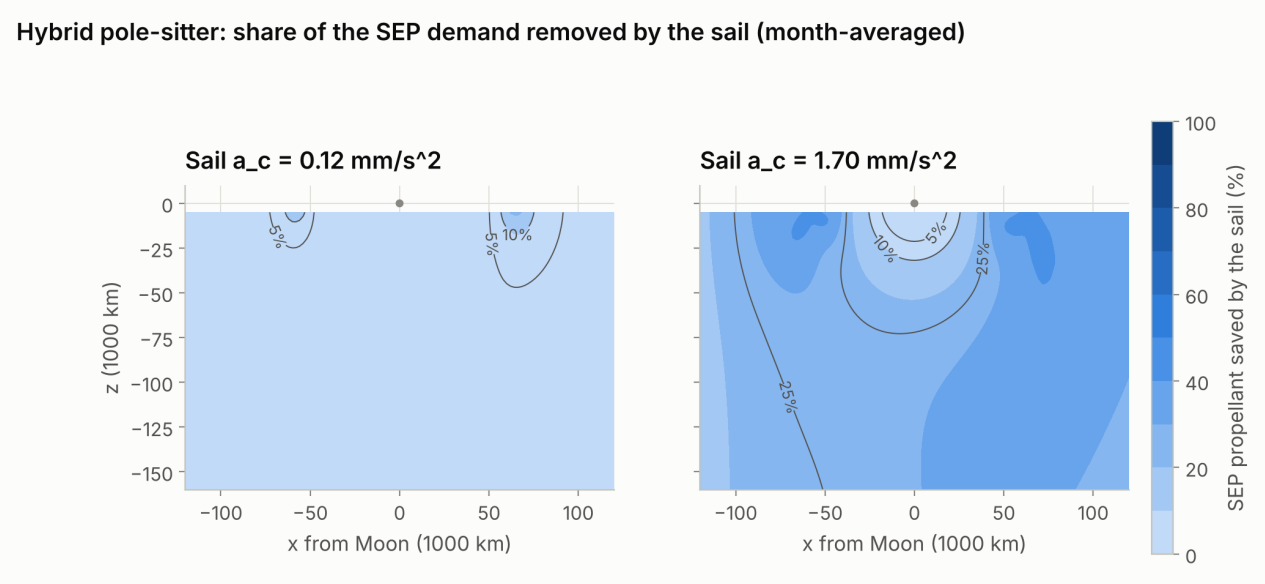


Figure 7.2 - Share of the SEP demand removed by a sail (month-averaged): ~0-10 % at 0.12 mm/s^2, ~10-40 % even at 1.7 mm/s^2.

- The requirement is dominated by lunar gravity along the polar axis; the sail's thrust lies near the Earth-Moon plane (the Sun is within 5 deg of it), almost perpendicular to what is needed. Hence the small savings in Figure 7.2.
- A static hover is therefore an SEP mission: Grebow, Ozimek & Howell needed a 500 kg spacecraft (450 kg propellant, 150 mN) for 387-554 days of pole-sitting, consistent with the 31-56 km/s per year found here.
- Sails make sense only on non-static, synodic-period 'pole-coverage' orbits that let the spacecraft drift with the Sun: Ozimek et al. found single-sail coverage needs kappa >= 0.58 mm/s^2 (elevations of only 4-7 deg near L1/L2) and ~1.7 mm/s^2 for 15-45 deg elevations; Wawrzyniak & Howell require >= 1.5 mm/s^2 for a 15 deg constraint.

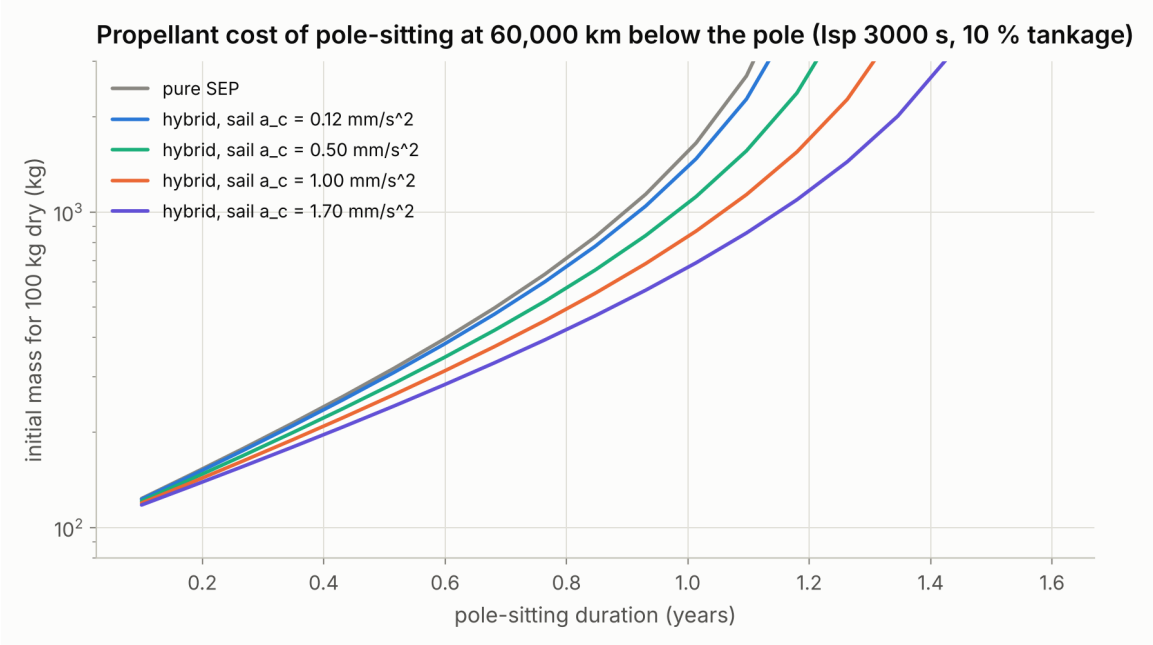


Figure 7.3 - Initial mass needed for 100 kg dry mass vs pole-sitting duration at 60,000 km below the pole (Isp 3,000 s).

Answer to Q4

Necessary: either (a) a sail with a_c >= ~0.6 mm/s^2 (low-elevation coverage) to ~1.7 mm/s^2 (high elevation) flying synodic-period coverage orbits - a total sail loading of ~5-15 g/m^2, i.e. 5-14x the CR-S performance and beyond the far-term CR-A (0.27 mm/s^2); or (b) high-Isp SEP supplying 31-56 km/s per year, which consumes 65-85 % of the mass per year at Isp 3,000 s. **Recommendation:** do not baseline a lunar pole-sitter for Cattle Run; provide polar coverage with two spacecraft in NRHO/L2-halo orbits (natural dynamics, sail station keeping) and revisit pole-sitting when sail loading below ~10 g/m^2 is demonstrated.

8. Q5 - The 'no travel' (in irons) region: sail vs hybrid

A sail cannot push toward the Sun. With a real film part of the momentum is absorbed or diffusely reflected, so the force can never tilt more than ~50-56 deg from the sunlight direction: the reachable accelerations form a 'bubble' touching the origin, and a cone around the sunward direction is completely unreachable.

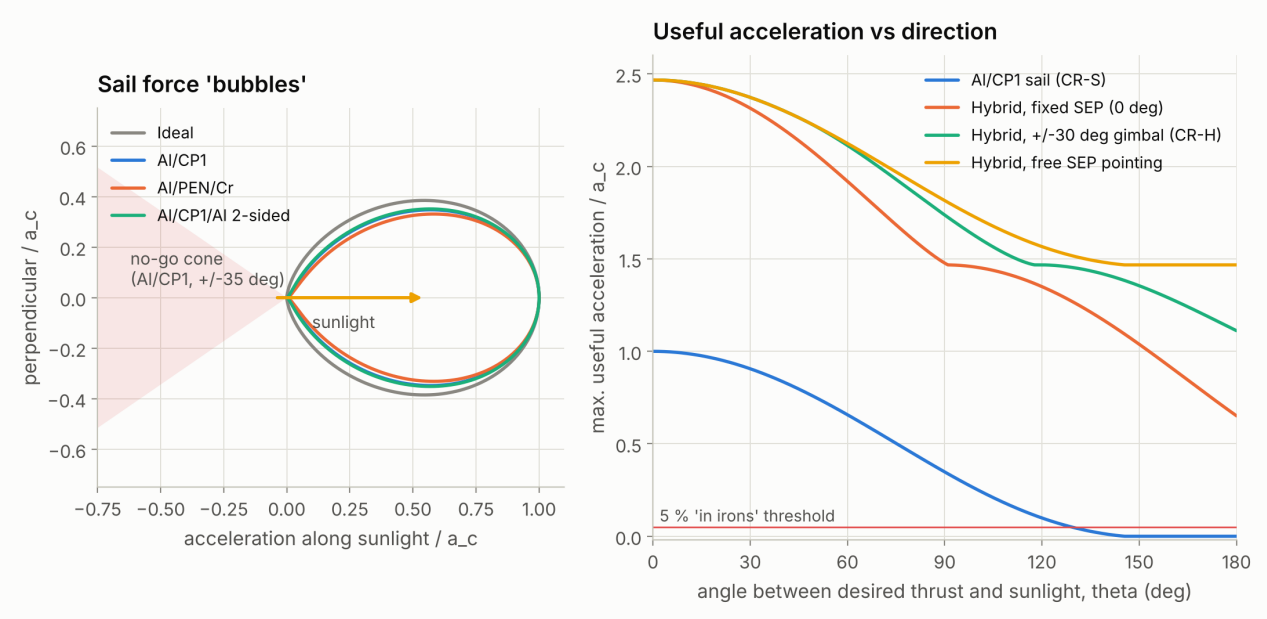


Figure 8.1 - Left: force bubbles of ideal and real films with the Al/CP1 no-go cone. Right: maximum useful acceleration (support function) vs direction for the sail and three hybrid variants.

Configuration	No thrust at all within (deg of sunward)	< 5 % a_c within (deg)	Toward Sun (x a_c)	Perpendicular (x a_c)
Ideal sail	1.0	41.0	0.00	0.38
Al/CP1 sail (CR-S)	34.5	50.8	0.00	0.35
Al/PEN/Cr sail (ACS3)	40.8	54.5	0.00	0.33
Hybrid, fixed SEP (0 deg)	0.0	0.0	0.65	1.48
Hybrid, +/-30 deg gimbal (CR-H)	0.0	0.0	1.11	1.74
Hybrid, free SEP pointing	0.0	0.0	1.47	1.82
Hybrid, small SEP (0.25 a_c, +/-30 deg)	0.0	0.0	0.14	0.58

Table 8.1 - In-irons cones and directional capability (SEP sized as CR-H unless noted).

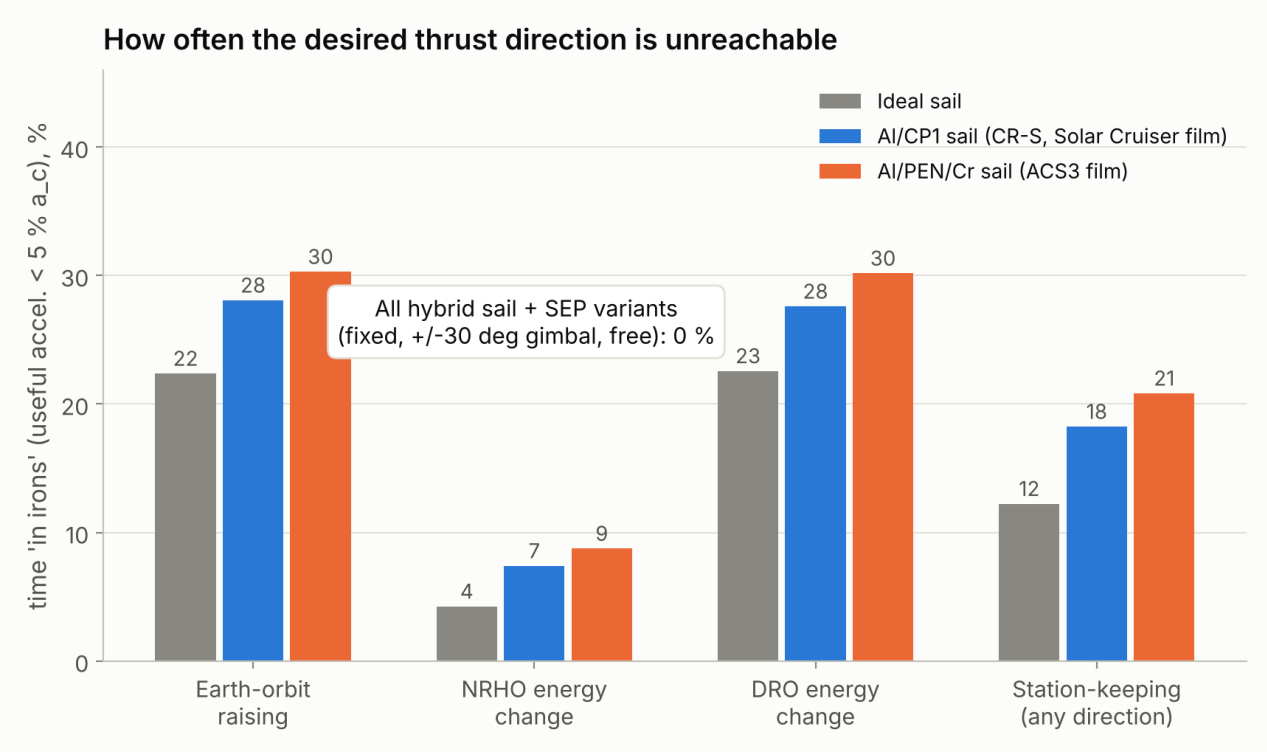


Figure 8.2 - Share of time the desired thrust direction is unreachable during typical manoeuvres (useful acceleration below 5 % of a_c).

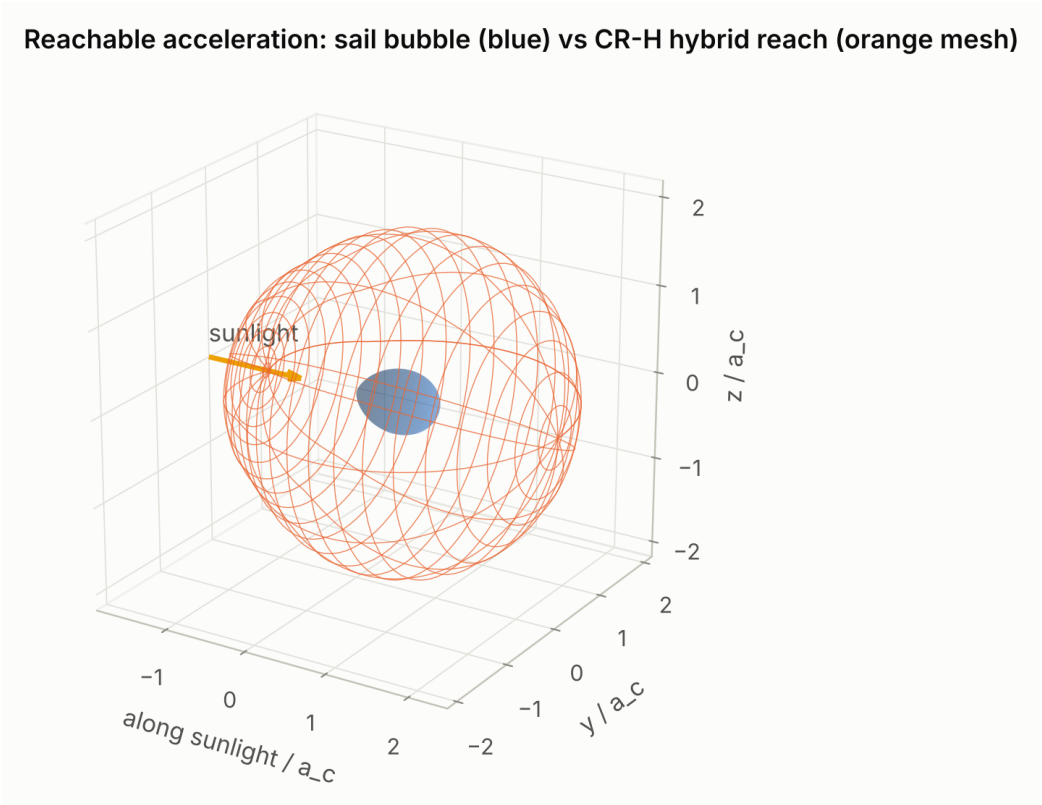


Figure 8.3 - 3-D reachable-acceleration sets: the sail bubble (blue) inside the CR-H hybrid's reach surface (orange).

Answer to Q5

Sail: a cone of 34.5 deg half-angle around the sunward direction is unreachable for Al/CP1 (40.8 deg for ACS3's Al/PEN/Cr), and thrust is below 5 % of a_c out to 50.8 deg - 18 % of all directions. Operationally the sail is in irons 28 % of the time when raising an Earth orbit, 28 % in a DRO, 7 % in an NRHO and 18 % for arbitrary station-keeping corrections. **Hybrid:** no in-irons region at all. Even a fixed (ungimballed) SEP gives 0.65 a_c sunward; CR-H's +/-30 deg gimbal gives 1.11 a_c.

9. Q6 - Field of view: fixed camera, antenna and arrays, with and without a gimballed bus

A sailcraft's attitude belongs to its propulsion: the sail normal follows the steering law. A body-fixed instrument therefore points wherever the thrust demand puts it. We simulated four months of operations in a 70,000 km DRO and the 9:2 NRHO, with sail attitudes from the locally optimal law for (a) random manoeuvre demands updated every 12 h and (b) energy-change steering, and counted how often each target is inside the instrument field. Free roll about the sail normal is assumed (the sail force does not depend on roll), and the membrane blocks +/-5 deg about its plane.

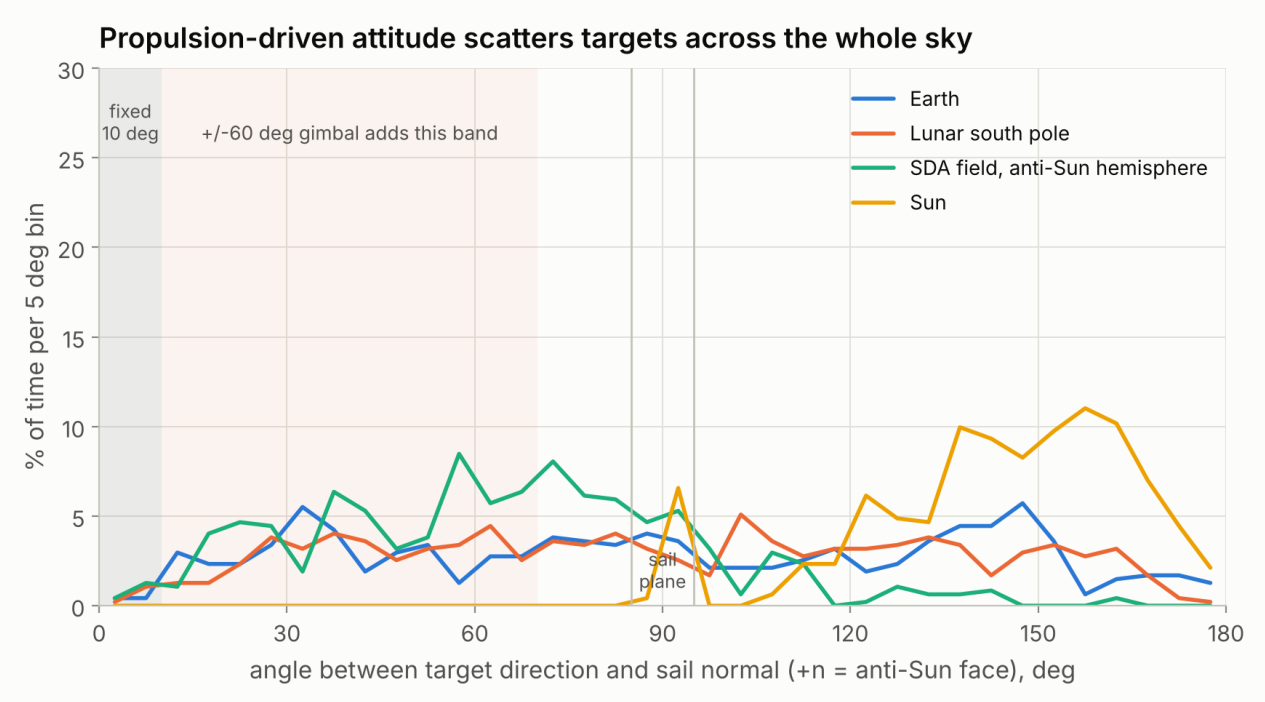


Figure 9.1 - Distribution of the angle between targets and the sail normal: propulsion-driven attitude spreads Earth, the lunar south pole and SDA targets across the sky.

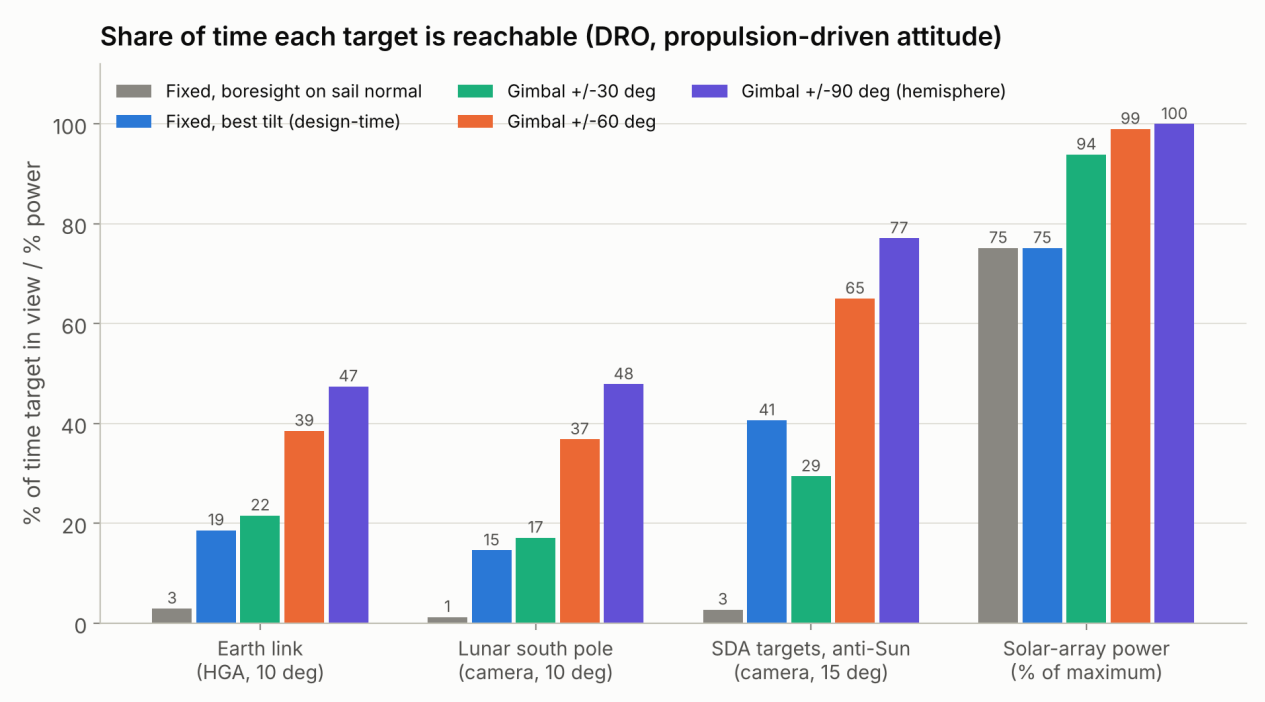


Figure 9.2 - Share of time each target is in view and mean solar-array power for fixed and gimballed buses (DRO, manoeuvre demand).

Case	Fixed	Fixed, best tilt	Gimbal +/-30 deg	Gimbal +/-60 deg	Gimbal +/-90 deg (hemisphere)
DRO 70k / maneuver: Earth	3 %	19 %	22 %	39 %	47 %
DRO 70k / maneuver: Lunar south pole	1 %	15 %	17 %	37 %	48 %
DRO 70k / maneuver: SDA field, anti-Sun hemisphere	3 %	41 %	29 %	65 %	77 %
DRO 70k / maneuver: Solar array power	75 %	75 %	94 %	99 %	100 %
DRO 70k / energy: Earth	5 %	10 %	19 %	32 %	39 %
DRO 70k / energy: Lunar south pole	3 %	14 %	19 %	34 %	37 %
DRO 70k / energy: SDA field, anti-Sun hemisphere	4 %	30 %	25 %	55 %	66 %
DRO 70k / energy: Solar array power	63 %	63 %	86 %	97 %	100 %
NRHO 9:2 / maneuver: Earth	3 %	16 %	17 %	36 %	47 %
NRHO 9:2 / maneuver: Lunar south pole	0 %	22 %	6 %	25 %	43 %
NRHO 9:2 / maneuver: SDA field, anti-Sun hemisphere	3 %	41 %	29 %	65 %	77 %
NRHO 9:2 / maneuver: Solar array power	75 %	75 %	94 %	99 %	100 %
NRHO 9:2 / energy: Earth	0 %	18 %	14 %	36 %	47 %
NRHO 9:2 / energy: Lunar south pole	0 %	23 %	10 %	28 %	47 %
NRHO 9:2 / energy: SDA field, anti-Sun hemisphere	4 %	39 %	25 %	63 %	74 %
NRHO 9:2 / energy: Solar array power	78 %	78 %	97 %	100 %	100 %

Table 9.1 - Access fractions for all simulated cases.

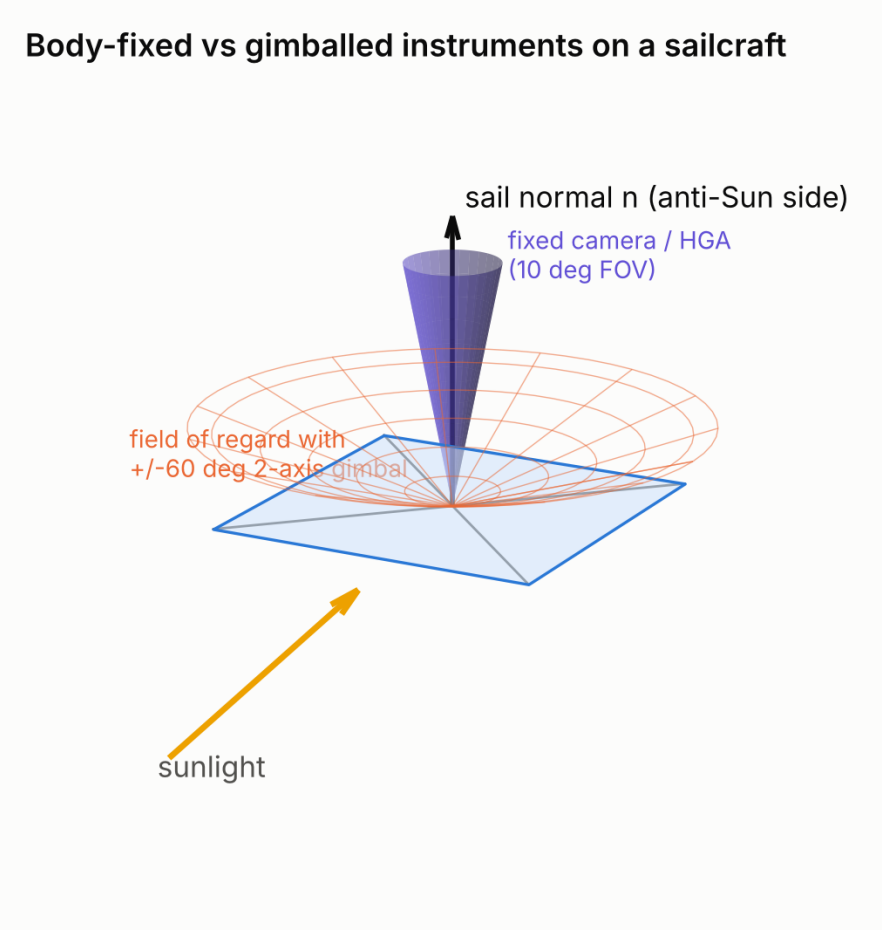


Figure 9.3 - Fixed instrument field (violet) vs field of regard of a +/-60 deg two-axis gimballed bus (orange).

Answer to Q6

Fixed bus: a 10 deg antenna or camera on the sail normal sees Earth 3 % and the lunar south pole 1 % of the time; the best design-time tilt raises this to 19 %/15 %. Fixed arrays average 75 % of peak power (zero when feathered). **Gimballed bus:** +/-30 deg recovers 94 % power; +/-60 deg gives 39 %/37 % access and 65 % for anti-Sun SDA fields. The sail itself caps any single-face instrument at ~50 %: carry antennas on both faces, or schedule pointing windows into the steering plan.

10. Q7 - Manoeuvrability and acceleration of the hybrid approach

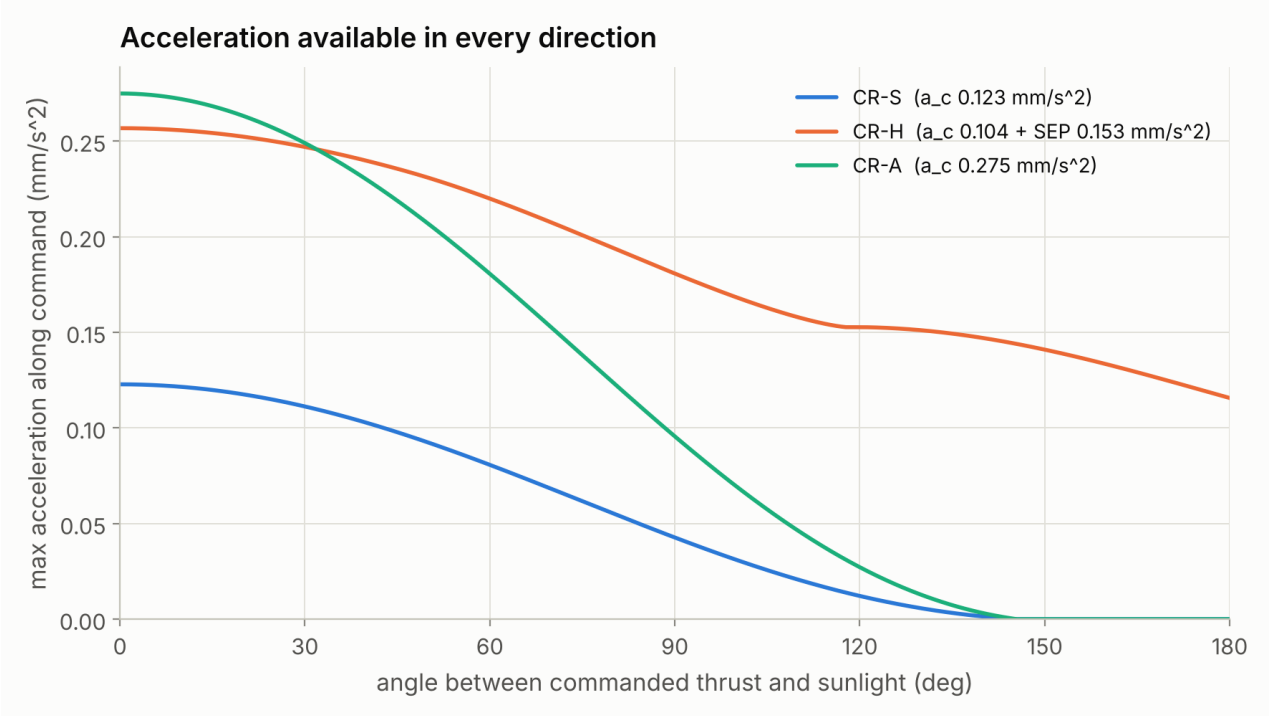


Figure 10.1 - Maximum acceleration along a commanded direction for CR-S, CR-H and the far-term CR-A.

Design	Away from Sun	Perpendicular	Toward Sun	Isotropic mean	dv per year, mean direction (km/s)
CR-S	0.123	0.043	0.000	0.048	1.5
CR-H	0.257	0.181	0.116	0.187	5.9
CR-A	0.275	0.096	0.000	0.108	3.4

Table 10.1 - Directional acceleration capability (mm/s^2). CR-H's SEP is propellant-limited to 0.69 km/s in total with 6 kg; the sail part is unlimited.

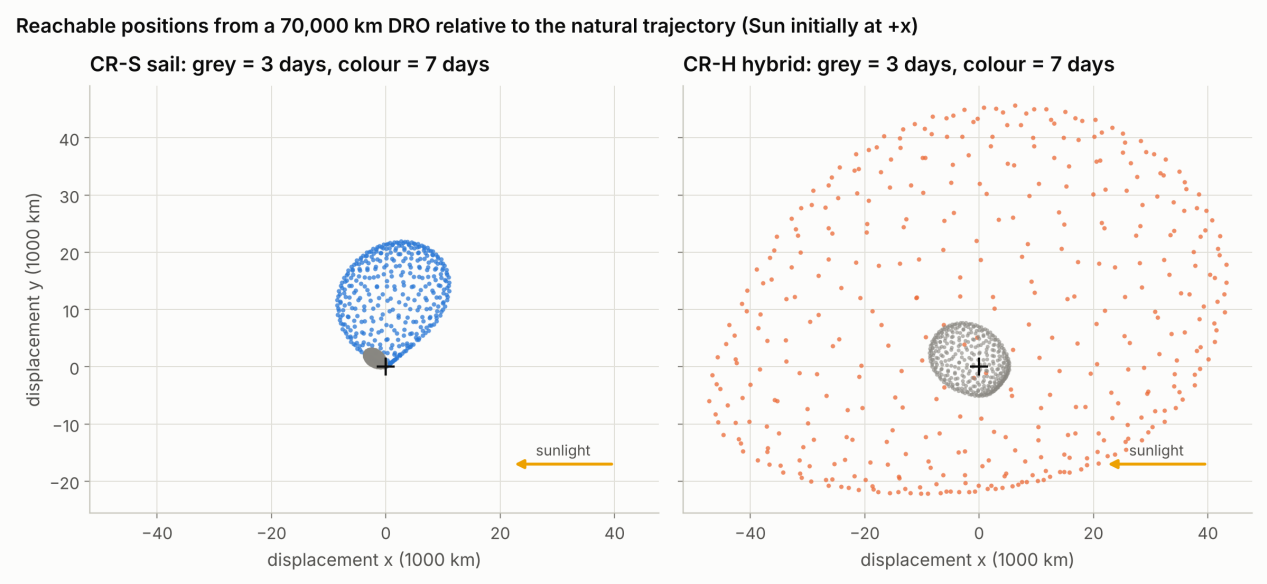


Figure 10.2 - Positions reachable from a 70,000 km DRO after 3 (grey) and 7 days (colour) with constant-direction steering commands, relative to the unpowered trajectory.

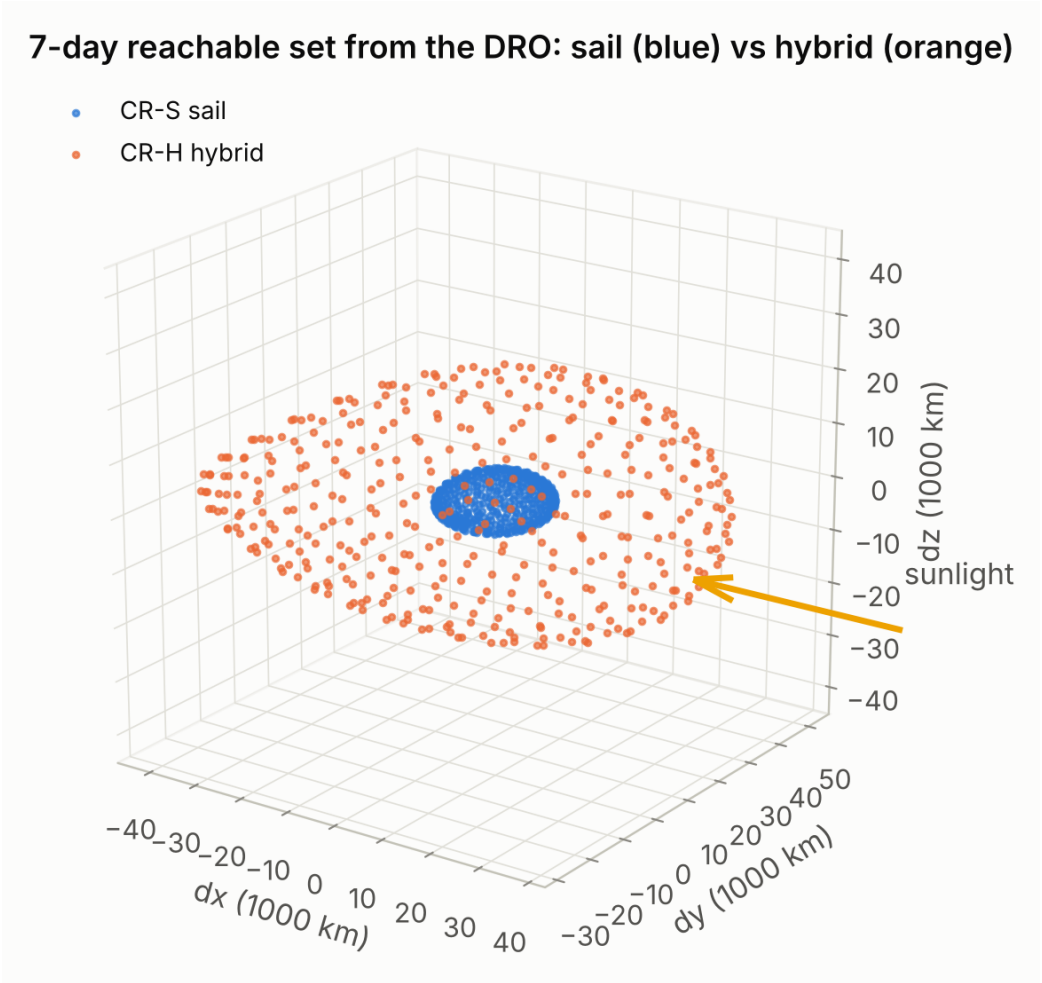


Figure 10.3 - 3-D view of the 7-day reachable sets.

- **Reach:** in 3 days the hybrid can displace 9,402 km, including 5,294 km toward the Sun, where the sail manages 2 km; after 7 days 48,878 vs 22,224 km.
- **Response:** a 10 m/s manoeuvre takes 0.94 d (away from Sun) to 2.71 d (perpendicular) with the sail and is impossible toward the Sun; the hybrid needs 0.45-1.00 d in any direction.
- **Mass:** the 1,653 m^2 sail system weighs ~33 kg, so a sail pays for itself against SEP-only only after ~3.6 km/s (sail-only) or ~6.7 km/s (hybrid) of cumulative manoeuvring - i.e. for long, manoeuvre-intensive constellation missions, exactly the Cattle Run use case.

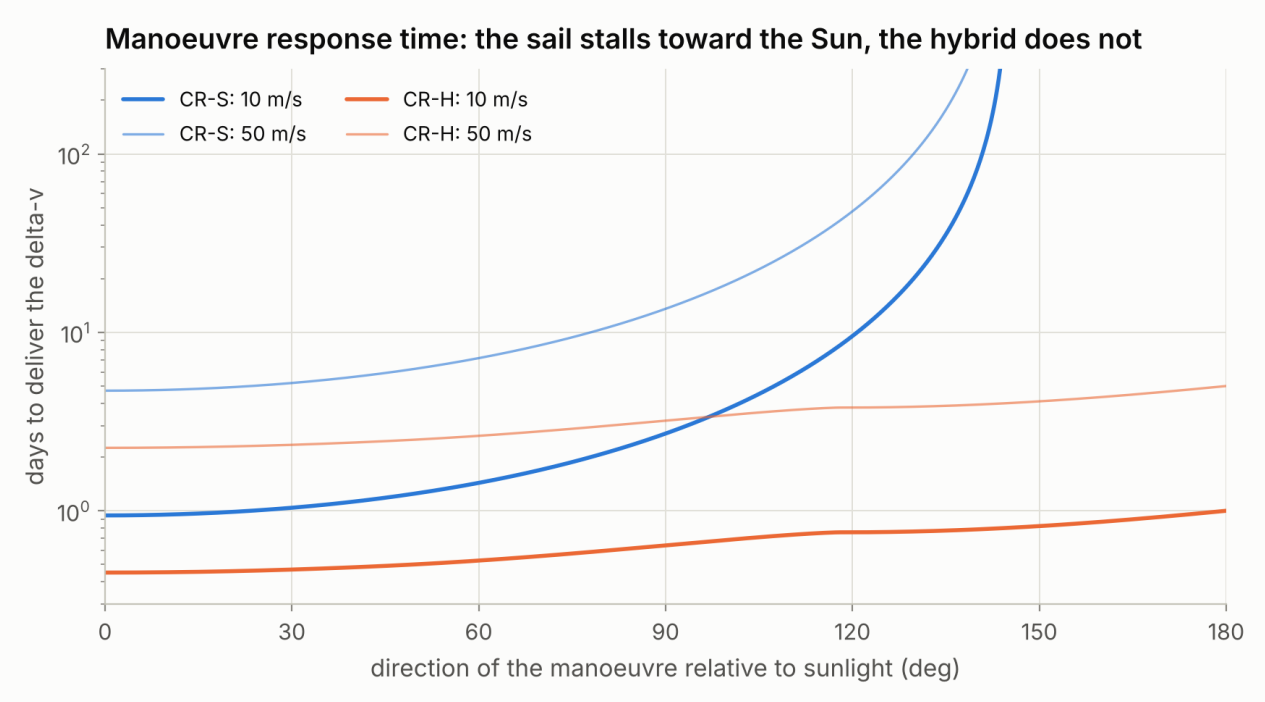


Figure 10.4 - Days to deliver 10 and 50 m/s vs manoeuvre direction.

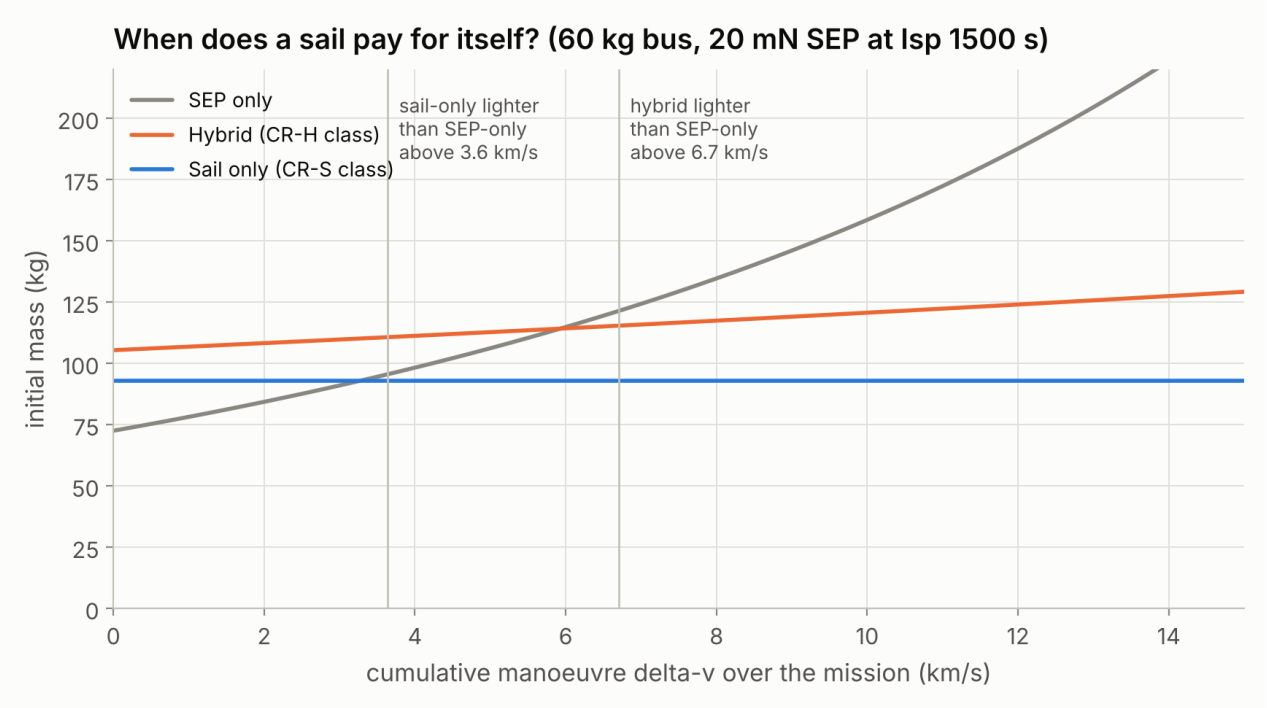


Figure 10.5 - Initial mass vs cumulative manoeuvre delta-v for SEP-only, hybrid and sail-only.

Answer to Q7

The hybrid is omnidirectional and roughly 2-4x as agile as the sail: 0.116-0.257 mm/s^2 in every direction (0.187 on average vs 0.048 for the sail), twice the 7-day reach and no stall toward the Sun. The SEP should be used tactically (in-irons gaps, rapid response, station-keeping precision) while the sail provides the bulk delta-v; with this split the 6 kg propellant load lasts for years.

11. Q8 - Lowest acceptable orbit for the below-GEO demonstrator

The demonstrator must show sufficient acceleration, accurate control and autonomous GNC. Below a certain altitude atmospheric drag swamps sunlight pressure, so orbit raising becomes impossible whatever the sail size: both forces scale with area/mass, and the criterion depends only on density (Mengali & Quarta):

$$f = \frac{\rho v^2}{2P} \leq \frac{2}{C_D} \quad \Longleftrightarrow \quad \text{maximum drag} \leq \text{maximum sail thrust}$$

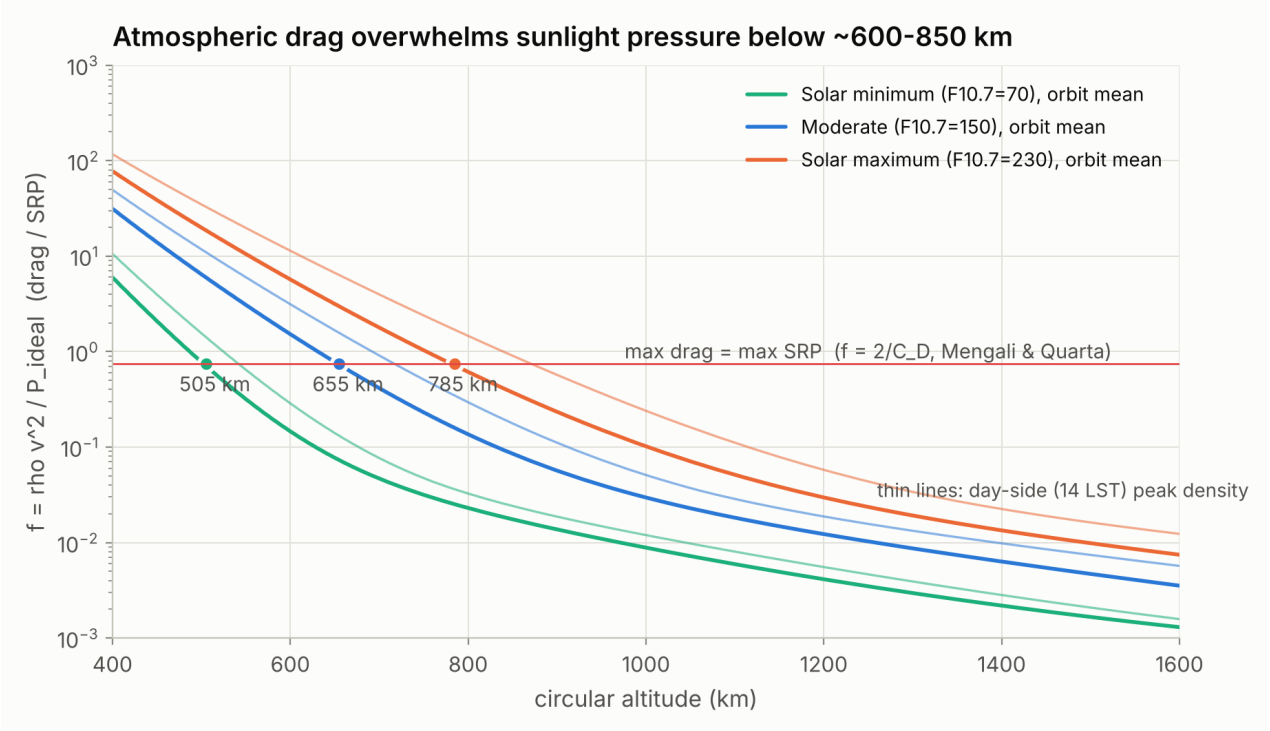


Figure 11.1 - Drag-to-sail-thrust ratio vs altitude for solar minimum, moderate and maximum activity (NRLMSIS 2.x).

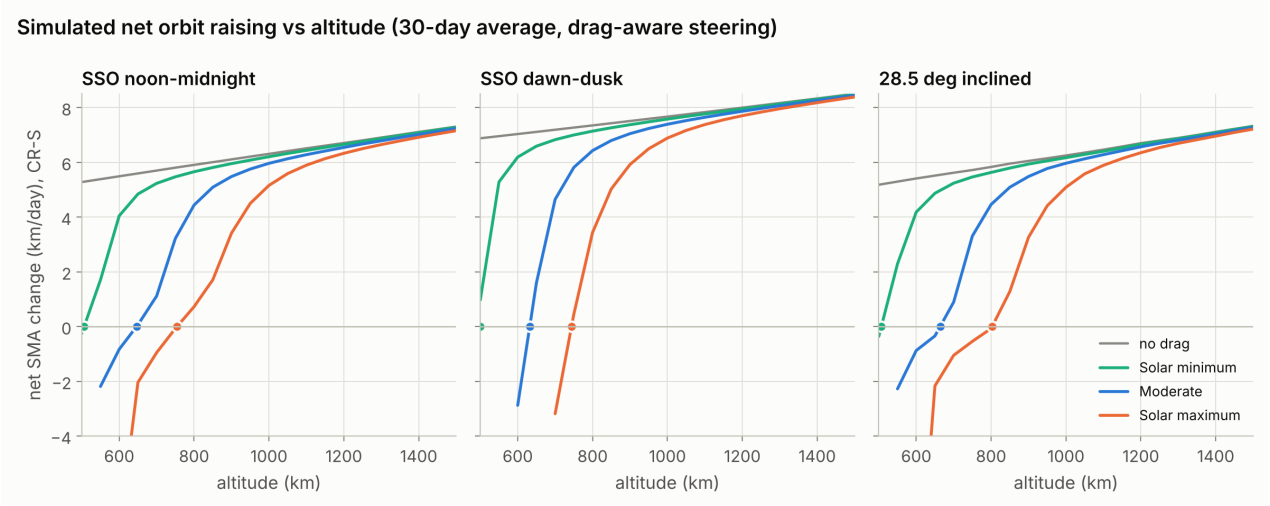


Figure 11.2 - Simulated 30-day net semi-major-axis change for CR-S in three orbit types (dots: break-even altitude).

Criterion	Solar min (F10.7 70)	Moderate (150)	Solar max (230)
Max drag = max thrust, orbit mean (km)	505	655	785
Max drag = max thrust, day side (km)	545	720	875
Simulated break-even: noon-midnight / dawn-dusk / 28.5 deg (km)	506 / 500 / 507	646 / 632 / 664	754 / 744 / 803
Drag <= 25 % of raising rate: noon-midnight / dawn-dusk / 28.5 deg (km)	650 / 550 / 600	800 / 750 / 800	1,000 / 900 / 1,000

Altitude (km)	Eclipse (beta 0)	AO fluence /cm^2/yr (moderate)	AO fluence (solar max)	Passive decay, CR-S deployed	Passive decay, ACS3
550	37 %	4.1e+20	1.4e+21	10 d	15 d
700	36 %	4.9e+19	2.3e+20	25 d	60 d
800	35 %	1.3e+19	7.6e+19	65 d	170 d
900	34 %	3.5e+18	2.6e+19	165 d	460 d
1000	33 %	1.0e+18	9.1e+18	365 d	1050 d

Table 11.1/11.2 - Altitude criteria; environment and passive decay (sail deployed, 50 % average projected area, F10.7 = 150) - a large sail is also an effective drag device for disposal.

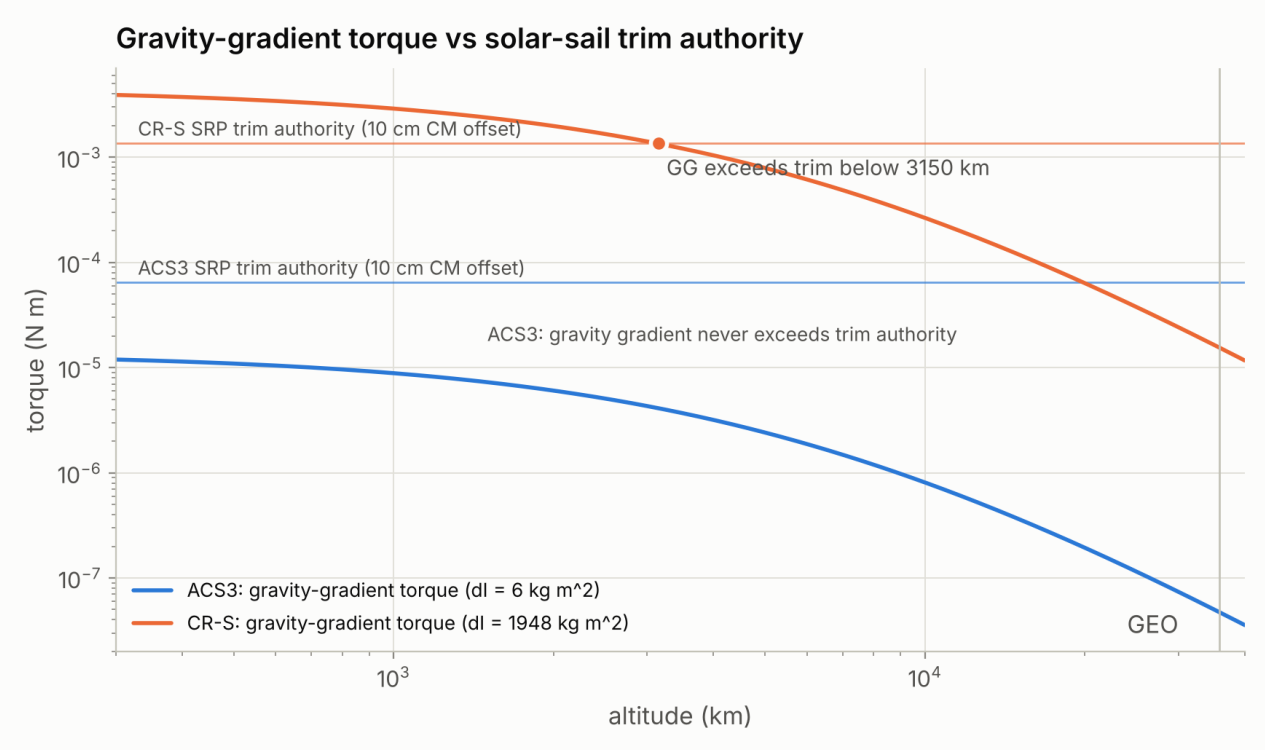


Figure 11.3 - Gravity-gradient torque (worst-case attitude) vs the solar-torque trim authority of a 10 cm centre-of-mass offset for ACS3 and the 1,653 m^2 CR-S sail.

- Drag sets a hard floor at ~632-664 km for moderate activity and ~744-803 km at solar maximum (2028 is the declining phase of Cycle 25; plan for F10.7 ~ 150 with excursions to 200+).
- Atomic oxygen erodes exposed polyimide at ~3e-24 cm^3/atom. For a ram-facing surface that is ~12 um per year at 550 km (moderate activity) - more than the 2.5 um film - 1.5 um at 700 km and 0.03 um at 1,000 km. The aluminised face protects the front, but defects and the bare back face are attacked; LightSail 2 showed AO damage at ~720 km.
- For a full-scale 1,653 m^2 sail, gravity-gradient torque exceeds the 10 cm CM-trim authority below ~3,150 km; a LEO demonstrator therefore needs GG-tolerant steering, magnetorquers and larger wheels, or a smaller demonstration sail (ACS3-class sails are never GG-limited).
- From a 550 km rideshare, CR-H can climb to 1,000 km with the sail stowed in 18 days using 2.1 kg of propellant - a strong argument for the hybrid (and for a retractable sail).

Answer to Q8

Absolute floor (drag = sail thrust): ~650 km for moderate activity, ~750-800 km at solar maximum. Lowest acceptable orbit for a convincing demonstration: ~1,000 km (900 km in a dawn-dusk SSO), where drag costs less than 25 % of the raising rate even at solar maximum, atomic-oxygen fluence is ~400x lower than at 550 km and a CR-S-class sail raises its orbit by ~222 km per month. Prefer a dawn-dusk SSO (no eclipses, ~20 % faster). If the demonstrator carries the full 1,653 m^2 sail, budget for gravity-gradient torques or fly at several thousand kilometres; otherwise demonstrate with a ~100-500 m^2 sail.

12. Design options and variables

The outline lists six design options and six variables. The simulator evaluates each; Table 12.1 summarises the trades and Figures 12.1-12.3 show the supporting analyses.

Option / variable	Benefit (quantified)	Cost / risk	Recommendation
Conventional sail (CR-S)	Propellant-free, unlimited delta-v (~1.5 km/s/yr mean-direction)	In irons 7-28 % of the time; no sunward thrust; attitude slaved to thrust	Baseline structure
Hybrid sail + SEP	No in-irons region; 2-4x agility; SEP climb from low rideshare	+20 kg (18 %); propellant; plume/sail interaction	**Adopt**
Gimballed vs fixed SEP	Sunward capability 1.11 vs 0.65 a_c; gimbal also trims CM (4e-3 N m)	Gimbal mass ~30 % of thruster; plume limits ~+/-45 deg	**Adopt +/-30 deg two-axis**
Retractable / re-deployable sail	Throttle: 11 % vs 2 % of demands met exactly; drag avoidance; protection in belts/conjunctions	+30 % deployer mass; mechanism wear; re-stow risk	Mature on the ground; fly on demo
Electrochromic films (RCD)	2 % area: 5.1e-04 N m pitch/yaw, 6.4e-05 N m roll; continuous dumping; wheel-free roll	Authority nulls near 60 deg incidence; durability; power	Adopt for roll + fine trim
Double- vs single-sided film	Efficiency 0.934 vs 0.908; either face usable (no flip slews)	Runs ~149 C hotter (139 C)	Single-sided CP1 (thermal margin)
Rigid booms vs spin/yo-yo	Spin saves ~11 kg on a 1,653 m^2 sail	Slow attitude changes (precession) - unsuitable for LEO raising (needs ~0.06 deg/s)	Rigid booms (agility)
Mass / size / material	a_c scales with A/m; 1 um film + 45 m booms reach 0.27 mm/s^2 (CR-A)	Larger sails raise GG torques and inertia	Grow a_c for later blocks
SEP thrust and Isp	20 mN at 1,500 s gives 0.15 mm/s^2; higher Isp stretches propellant	Power (~350 W) and arrays	Hall 15-25 mN class
Gimbal range	+/-30 deg: 94 % power; +/-60 deg: 2x instrument access	Mass, harness, stiffness	+/-30-60 deg

Table 12.1 - Design option trade summary.

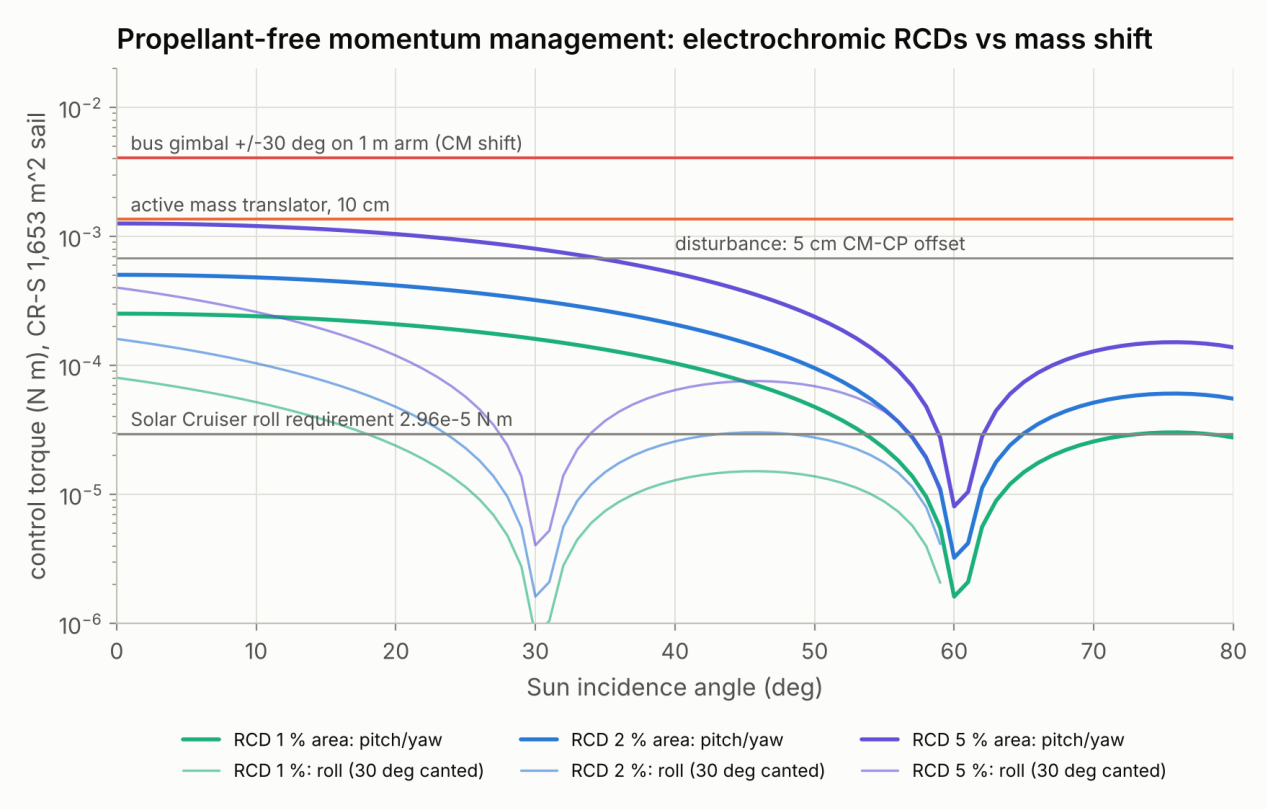


Figure 12.1 - Propellant-free momentum-management torque for the 1,653 m^2 sail: RCD panels (1/2/5 % of area) vs active mass translator and bus gimbal, against disturbance and Solar Cruiser's roll requirement.

A 5 cm offset between centre of mass and centre of pressure on CR-S accumulates 59 N m s per day - far beyond small-satellite wheels - so continuous trim by mass shift (AMT or bus gimbal) is mandatory and RCDs are the natural roll actuator (Inness 2023; Heaton 2023).

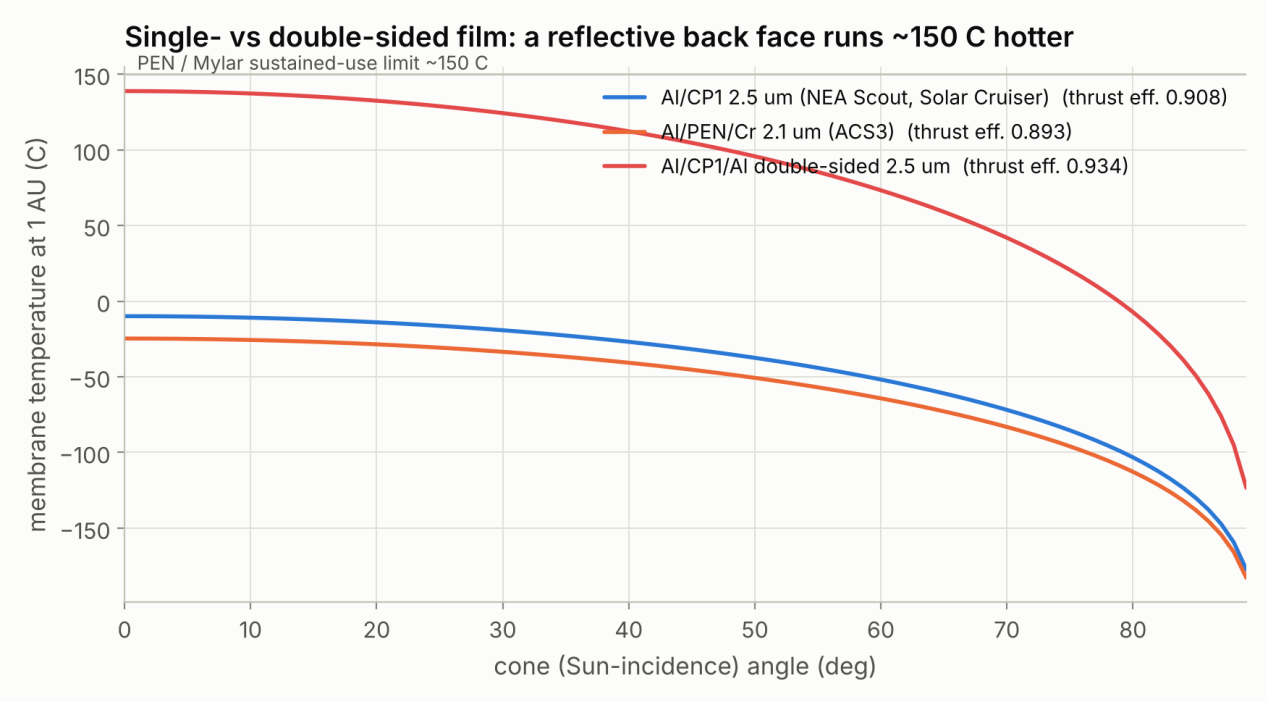


Figure 12.2 - Membrane temperature vs incidence: a double-sided reflective film cannot radiate from its back face and runs ~150 C hotter.

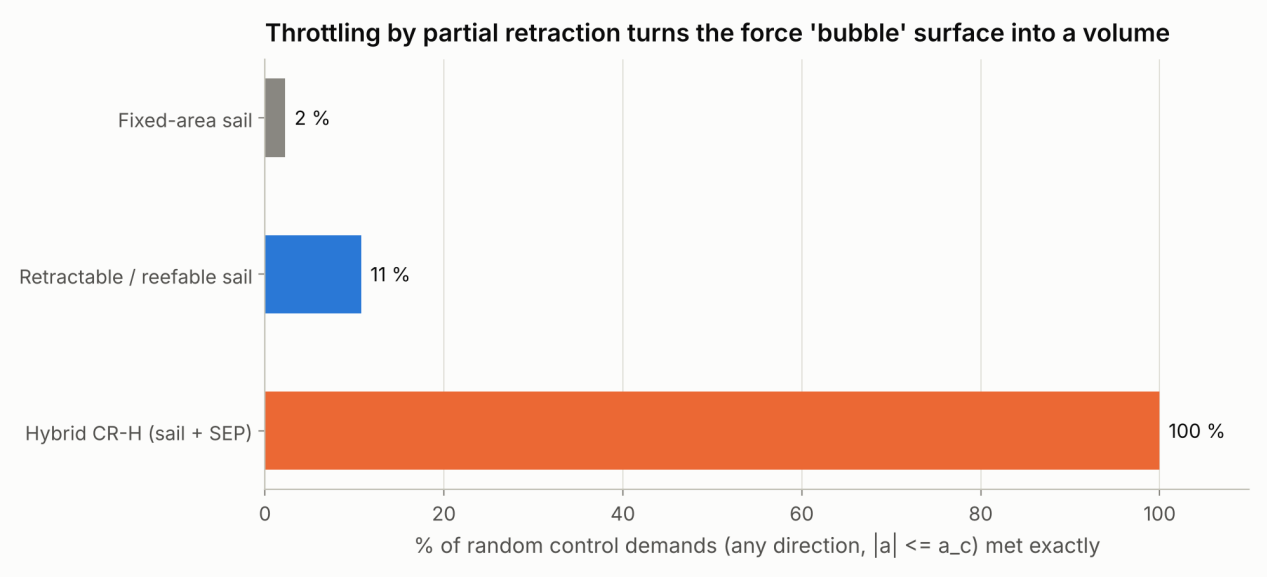


Figure 12.3 - Share of arbitrary control demands met exactly: fixed-area sail, retractable (reefable) sail, hybrid.

13. Additional questions that should be addressed

The outline asks to identify further questions. The analyses surfaced the following, roughly in priority order for the proposal:

#	Question	Why it matters / first evidence
A	How long do the film and coatings survive at the demo altitude and through the radiation belts?	AO fluence drops ~400x from 550 to 1,000 km; spiral-out crosses the inner belt for months; LightSail 2 degradation
B	What collision and conjunction risk does a 1,653 m^2 sail carry at ~1,000 km, and how is it screened?	Large area-to-mass objects; continuous-thrust ephemeris uncertainty (ACS3 provides predictive ephemerides to CARA)
C	How is orbit determination done with continuous, attitude-dependent SRP?	Force/torque model calibration from flight data (Diedrich 2023); GPS in LEO, optical/radiometric in cislunar space
D	What autonomy architecture guarantees safe feathering, momentum management and fault recovery?	Locally optimal laws are cheap on board; certification of autonomous GNC is a DARPA objective
E	Can the attitude system handle LEO gravity-gradient and aerodynamic torques for a large sail?	GG exceeds 10 cm CM-trim authority below ~3,150 km for 1,653 m^2
F	How do sail billow, wrinkles and boom thermal deflection change force and torque?	NEA Scout and LightSail 2 boom/sail deformation; RCD authority depends on incidence
G	Can the sail and booms survive lunar and Earth eclipses (thermal snap, contraction loads)?	NEA Scout booms were not designed for lunar-eclipse contraction loads
H	How should a constellation be deployed and phased from one launch?	Sail phasing in DRO/NRHO takes weeks; L4/L5 slots take 2-3 months (Section 6)
I	Where do antennas go when the sail blocks half the sky?	Single-face instruments are capped near 50 % access (Section 9)
J	Does SEP plume impingement contaminate the film, and what gimbal limits follow?	Constrains hybrid gimbal range
K	What is the station-keeping cost in the full ephemeris model with solar gravity and eclipses?	CR3BP results here; BCR4BP option is implemented for follow-on runs
L	How is end of life handled?	Sail as drag device in LEO (decay < 1 yr below 1,000 km); heliocentric escape from cislunar space

14. Recommended concept and next steps

- **Architecture:** hybrid sailcraft (CR-H class): Solar Cruiser-heritage 1,653 m^2 single-sided CP1 sail on rigid booms, RCD panels for roll, two-axis gimballed bus (+/-30-60 deg) carrying a 15-25 mN Hall thruster, arrays and instruments; antennas on both faces.
- **Demonstrator (Phase 1):** ~1,000 km dawn-dusk SSO, or a 550 km rideshare plus an 18-day SEP climb with the sail stowed. Success criteria: measured a_c within 10 % of prediction, net raising of ~5-7 km/day, autonomous steering and momentum management without magnetorquers for a defined period, safe feathering demonstrations.
- **Cislunar operations (Phase 2):** launch toward the lunar plane (28.5 deg) to GTO/GEO or TLI; slots in DRO, NRHO and L1/L2 halos with L4/L5 sentinels; polar coverage by paired NRHO spacecraft instead of a pole-sitter.
- **Simulator next steps:** trajectory optimisation (collocation) for the Q3 legs, full ephemeris and BCR4BP station-keeping with a sail-aware controller, slew-rate-limited attitude dynamics with RCD/AMT actuators, sail shape uncertainty, and GPU-scale Monte Carlo of autonomy fault cases.

15. Assumptions and limitations

- Flat sail with nominal optical properties; no degradation, billow or wrinkle losses (real performance typically 5-15 % lower).
- Attitude changes are instantaneous (no slew-rate or momentum limits) in trajectory runs; agility is treated separately.
- Orbit-averaged Earth-orbit dynamics omit lunisolar perturbations (0.1 % error after 60 days in raised GTO; larger near lunar distance).
- Cislunar dynamics use the circular restricted three-body problem with a rotating Sun line in the Earth-Moon plane; transfer times are energy-change plus natural-coast estimates, not optimised trajectories (expect +/-30 %).
- Drag uses NRLMSIS 2.x equatorial densities with local-time variation, C_D = 2.2 on projected area and a 1 % minimum (edge-on) area; latitude and seasonal variations are averaged out.
- Mass models are calibrated on ACS3 and Solar Cruiser data; SEP sizing follows Ceriotti & McInnes (25 kg/kW, 70 W/kg arrays).
- Field-of-view results assume free roll about the sail normal and simple membrane obstruction.

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Appendix A - Reproducing the results

All numbers and figures in this report are produced by the code in **simulator/**:

Command	Effect
pip install numpy scipy matplotlib pymsis (torch optional)	Dependencies; torch enables the GPU backend
python simulator/run_all.py	Runs Q1-Q8 and the design trades; writes results/*.json and figures/*.png (~10-15 min on 4 CPU cores)
CISLUNAR_BACKEND=torch python simulator/run_all.py	Same with PyTorch tensors (CUDA when available)
python simulator/run_all.py q3_cislunar_transfers	Re-run a single analysis
python reports/build_report.py	Rebuild this PDF from results and figures
node reports/build_deck.js	Rebuild the companion PowerPoint briefing

A.1 Key equations

$$P_{SEP} = \frac{T\, I_{sp}\, g_0}{2\,\eta}, \qquad \Delta v = I_{sp}\, g_0 \ln \frac{m_0}{m_f}$$

$$\tau_{GG} = \frac{3\mu}{2r^3}\,|I_n - I_t|\,\sin 2\theta, \qquad \tau_{trim} = F_{sail}\,\delta_{CM}$$

SEP power uses efficiency eta = 0.55 (thruster and PPU); the gravity-gradient torque uses the worst-case attitude (theta = 45 deg) and the difference between the sail's normal and in-plane moments of inertia from the mass model (membrane plus diagonal booms).